

Introduction: Contents

- Introduction
- Design Silicon Semiconductor manufacturing process
- Sculpting the Silicon CMOS on 180 nm Process Feature
- VLSI Fabrication

Introduction: Process Technologies

Operating two manufacturing facilities:

1. 200 mm Fab:

0.18 μm technology node

Core CMOS

Mixed signal

Analog

CCD image sensors



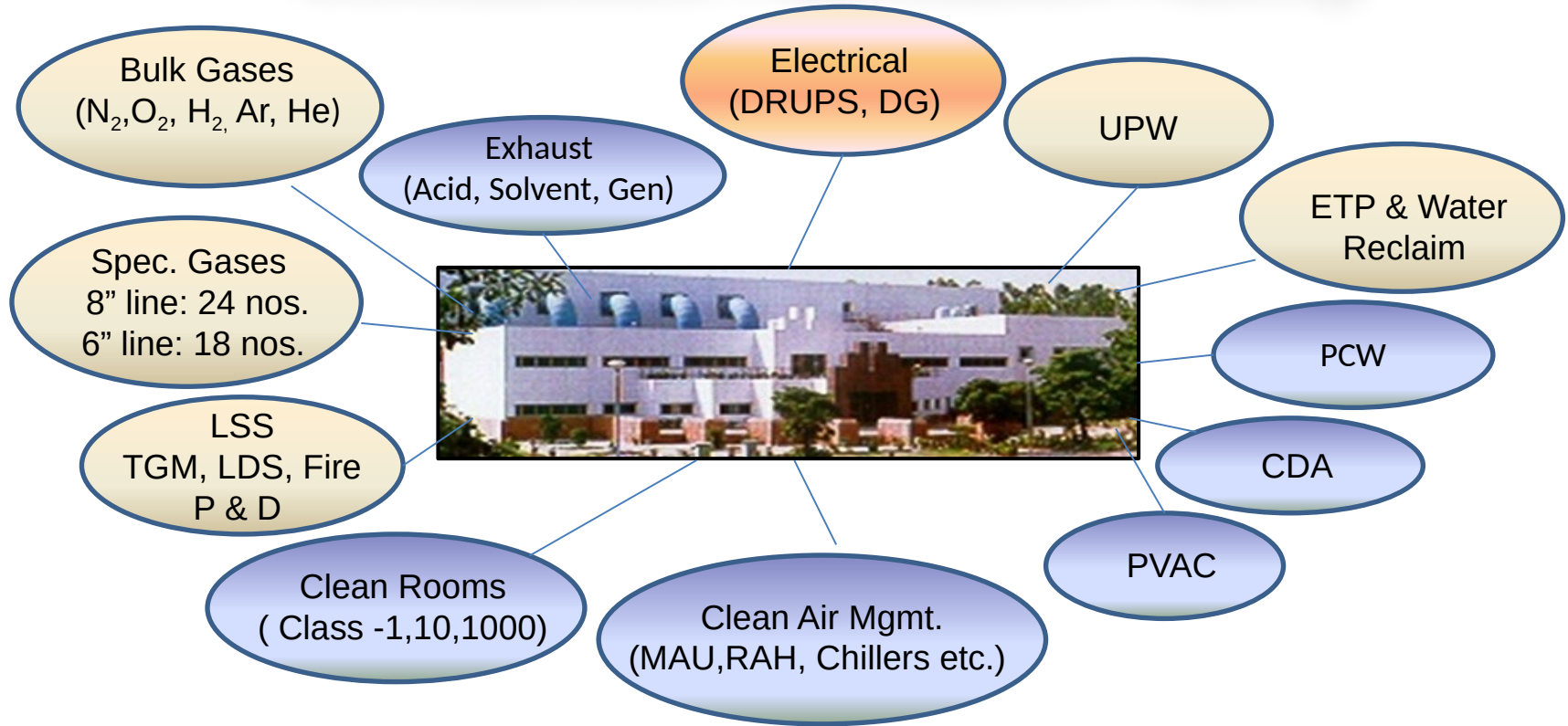
2. 150 mm Fab:

Dedicated for MEMS and Being Upgraded for III-V Compound

Semiconductor Based Devices

Litho resolution of 0.8 μm

Introduction: Fabrication Facility



Existing Infrastructure : For Fabrication Facility

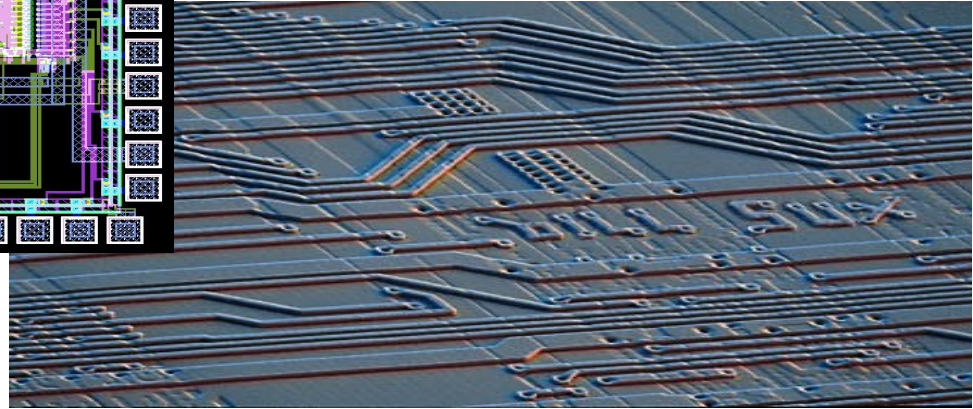
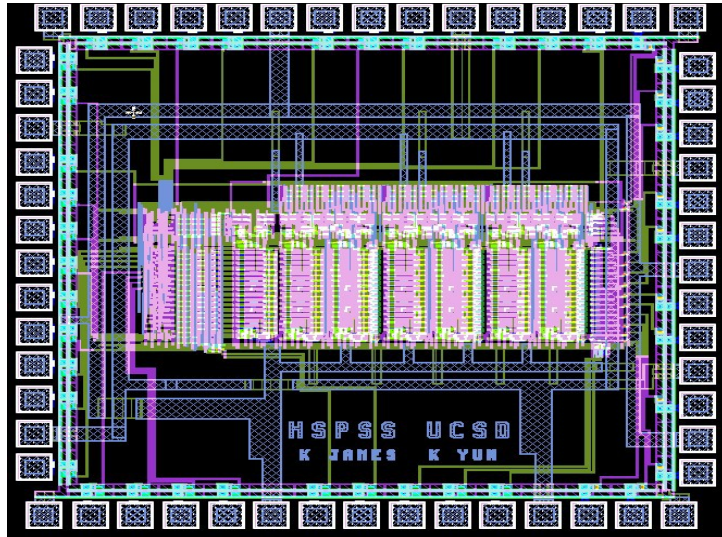
- **Clean Rooms (Class: 1,10,100,1K & 10K) : Area ~ 2100 sq. mt.**
Temp. ($21 \pm 0.5^{\circ}\text{C}$), RH ($40 \pm 2.5\%$), Δ Pr. (12 ± 6 Pa)
- **Air management Systems:**
 - Make-up Air Unit (MAU) : 50 K CFM (8"), 45K CFM (6")
 - Recirculation Air Handlers : 900K CFM
 - MT (8 ± 0.5 deg C) Centrifugal Chillers : 500 TR x 4 no.
 - LT (4 ± 0.5 deg C) Screw Chillers : 400 TR x 3 no.
- **Bulk gases ($\text{N}_2, \text{O}_2, \text{H}_2, \text{Ar}$ & He)**
- **Specialty Gases (34 nos.)**
- **Ultra Pure water Plant (Capacity 110 gpm / 180 gpm)**
- **Electrical Systems**
 - Sub-stations : 66/11KV and 11KV/415 V (3 No., with 9 Transformers)
 - Captive power : DRUPS (3x 1670 KVA), Static UPS (130 KVA),
DG Sets (2 x 1250 KVA + 1x 625 KVA)
- **Effluent treatment and Air abatement systems**
- **Life Safety Systems : TGM, Fire suppression & Detection systems**



Introduction: Process Technology

Module	Typical CMOS process flow
Well	Retrograde
Isolation	STI
GOX	Nitrided oxide
Gate	Dual poly
Spacer	Combination of Nitride & Oxide
Drain extension	Doping comparable to Drain
Contact	Salicide(CoSiO ₂)
Contact/Via fill	W plug
No of Metal	05-06
Planarization	CMP
IMD	USG
Passivation	Low stress Oxide/Nitride

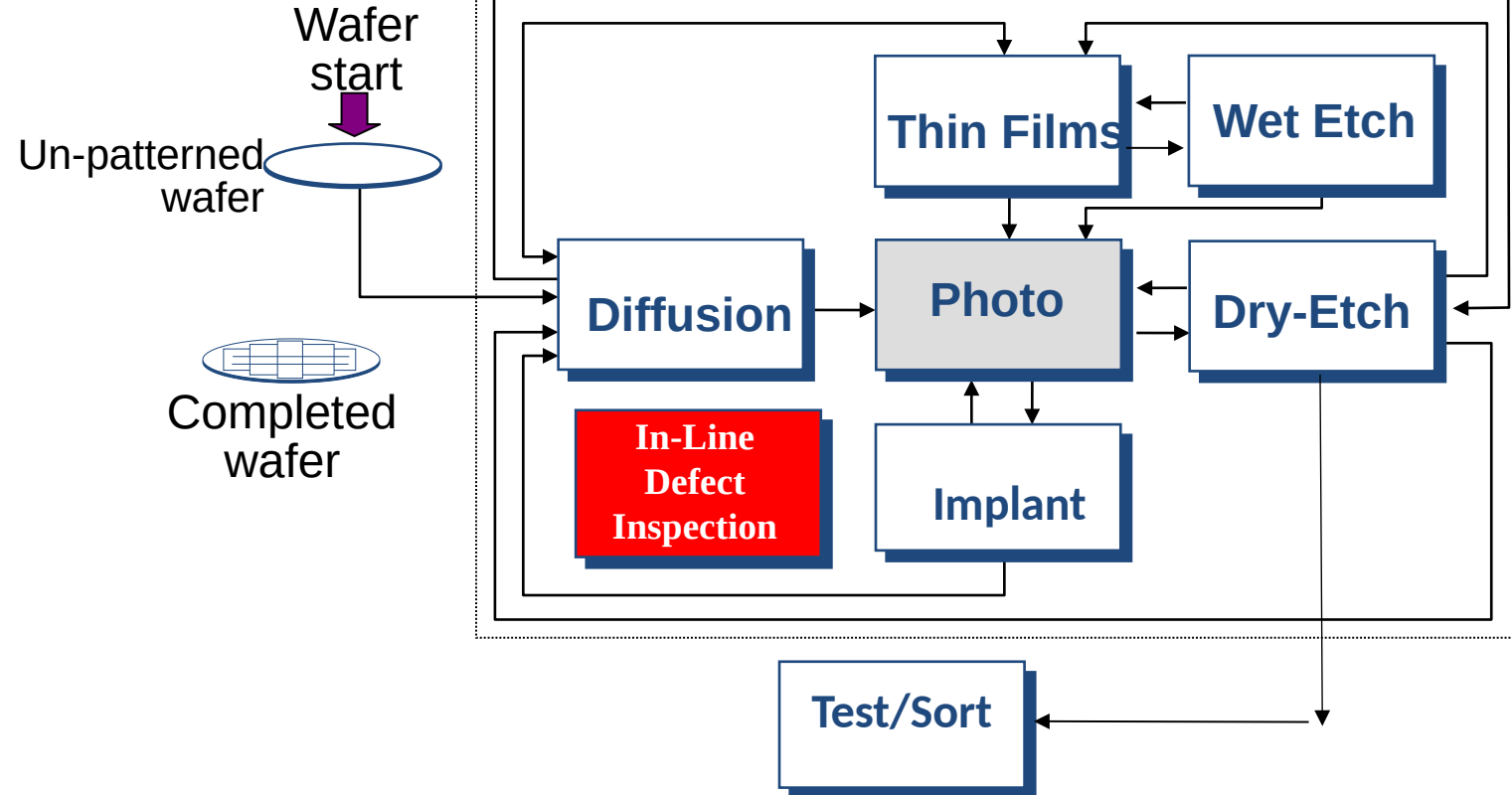
From Design to Reality



From Design to Reality

- Interface between the circuit designer and process engineer
- Guidelines for constructing process masks
- Unit dimension: minimum line width
 - scalable design rules: lambda parameter
 - absolute dimensions: micron rules
- Rules constructed to ensure that design works even when small fab errors (within some tolerance) occur
- A complete set includes
 - set of layers
 - intra-layer: relations between objects in the same layer
 - inter-layer: relations between objects on different layers

Wafer fabrication



Various Modules of 180 nm CMOS Process Technology

- Shallow Trench Isolation (STI)
- Retrograde Wells
- Gate Oxidation and Poly
- Shallow Source Drain (LDD) Extensions
- Contact Source / Drain Regions
- Silicided S/Ds and Polycides
- Contacts, Interconnects and Passivation

Starting Point:

Pure silicon 8 inch wafer



Silicon Wafer

Laser Marking

While transferring the wafers from Wafer Bank to Wafer Lot Box, Lot Identification Number along with Wafer Numbering on Wafers is done by LASER MARKING. All the processing is being done in class-1 Clean Room environment.

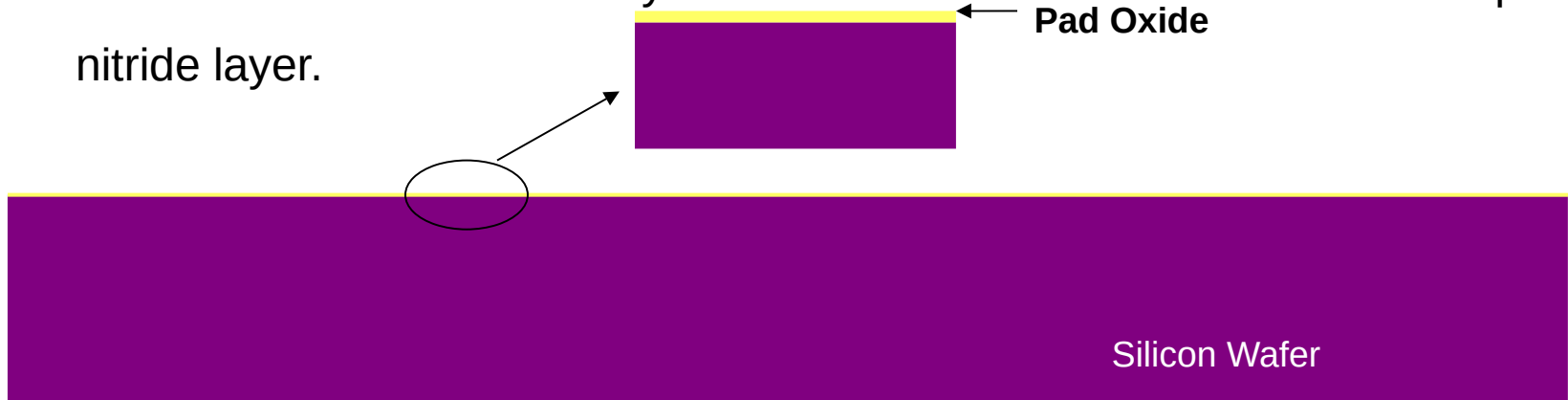


Silicon Wafer

Shallow Trench Formation

Furnace Grown Oxide

Grow Pad Oxide: A very thin layer of silicon dioxide (SiO_2) is grown on the surface by reacting silicon and oxygen at high temperatures. This will serve as a stress relief layer between the silicon and the subsequent nitride layer.



Shallow Trench Formation

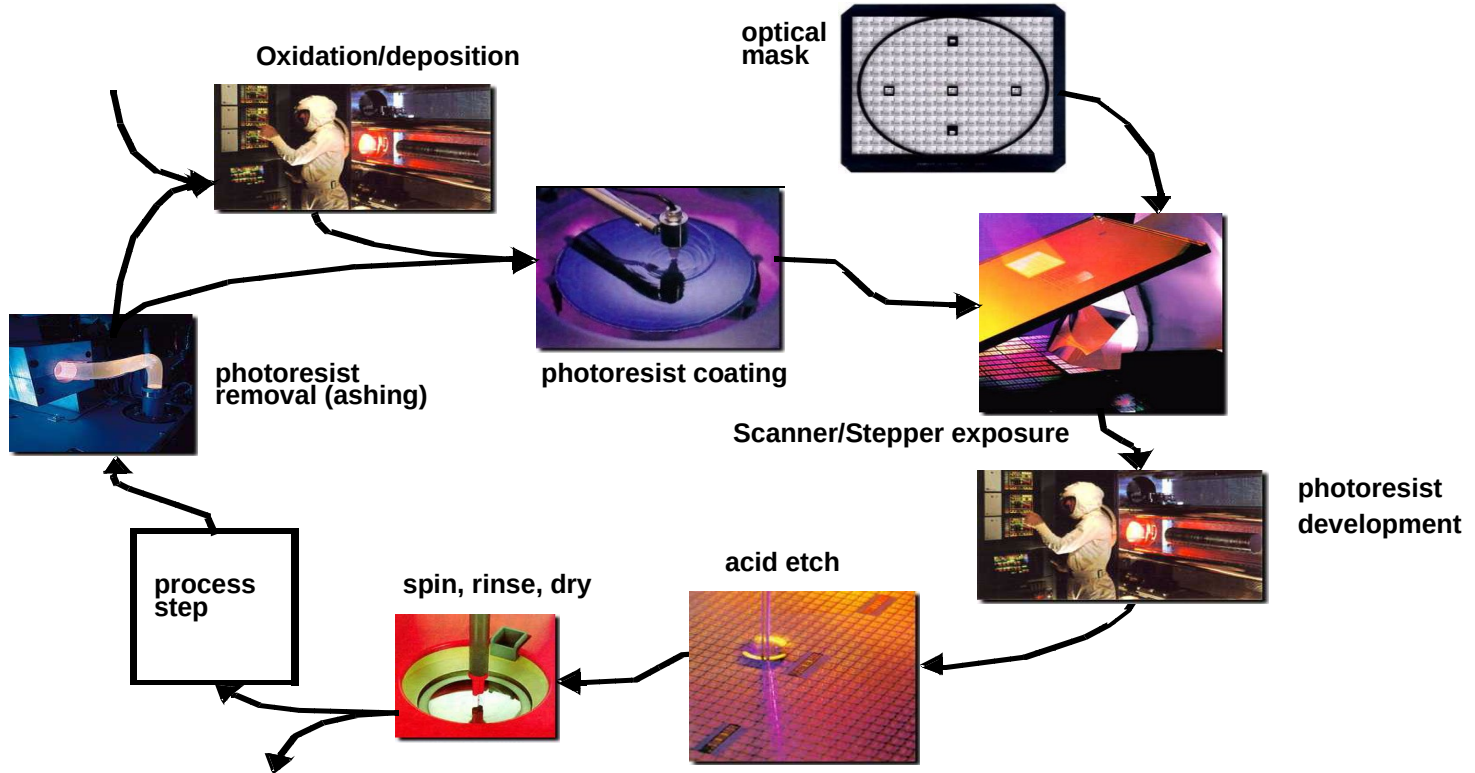
Furnace Nitride Deposition

Deposit Silicon Nitride: A layer of silicon nitride (Si_3N_4) is deposited using Chemical Vapor Deposition (CVD). This will serve as a polish stop layer during trench formation.



Photolithography Process Cycle

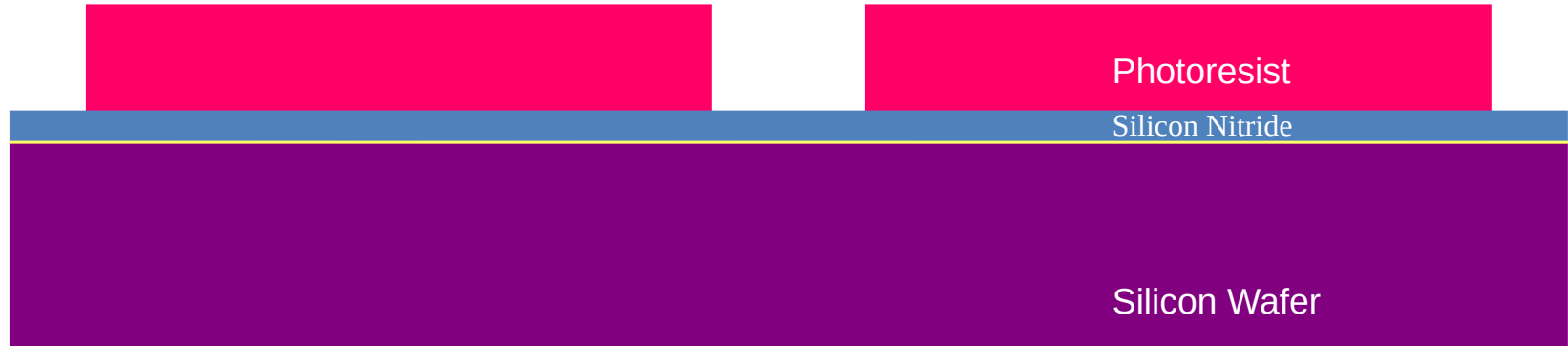
Typical operations in a single photolithographic cycle (from [Fullman]).



Shallow Trench Formation

DUV Patterning

Pattern Photoresist for Definition of Trenches: One of the most critical patterning steps in the process. Resist is spun, exposed, and developed.

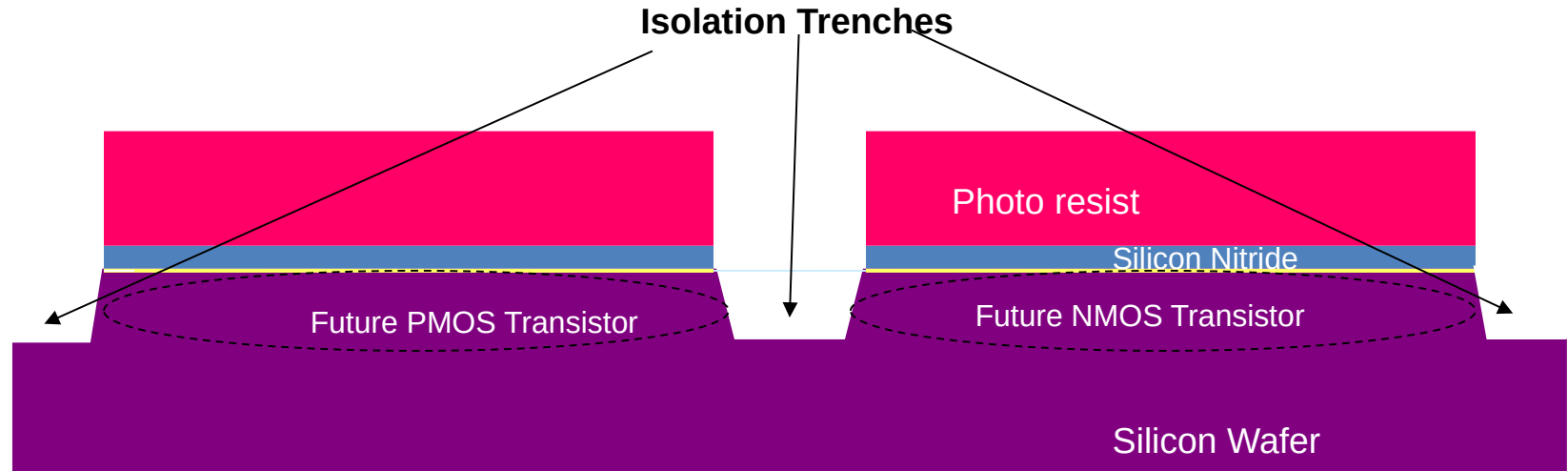


Shallow Trench Formation

Dry Etch

Etch Nitride and Pad Oxide: A reactive ion etch (RIE) utilizing fluorine chemistry is used.

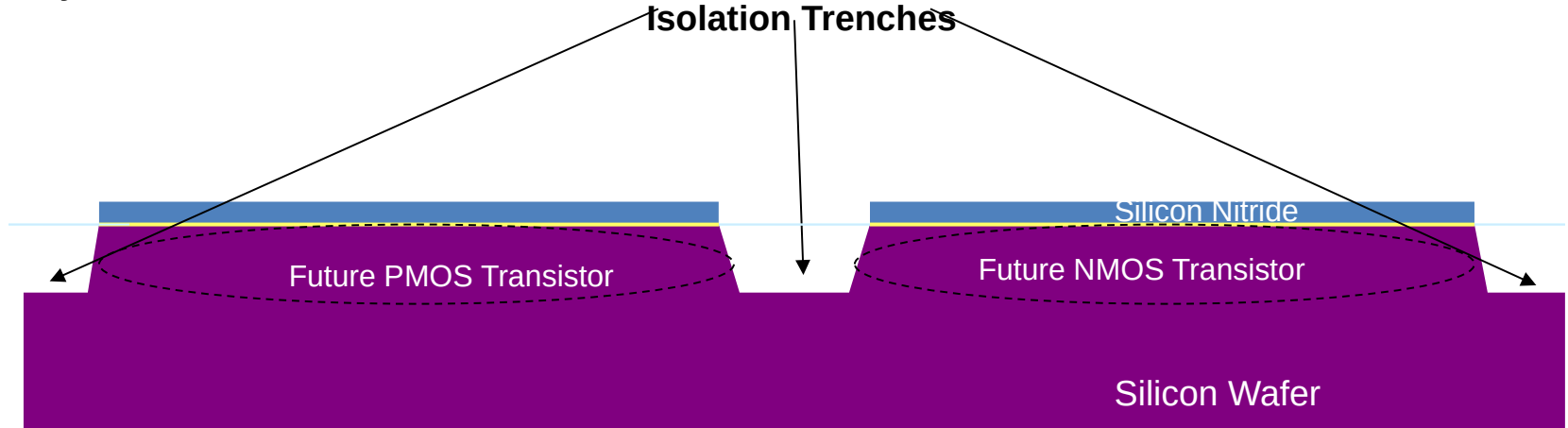
Etch Trenches in Silicon: A reactive ion etch (RIE) utilizing chlorine chemistry is used. Defines transistor active areas.



Shallow Trench Formation

Dry Etch: Photo Resist Ashing

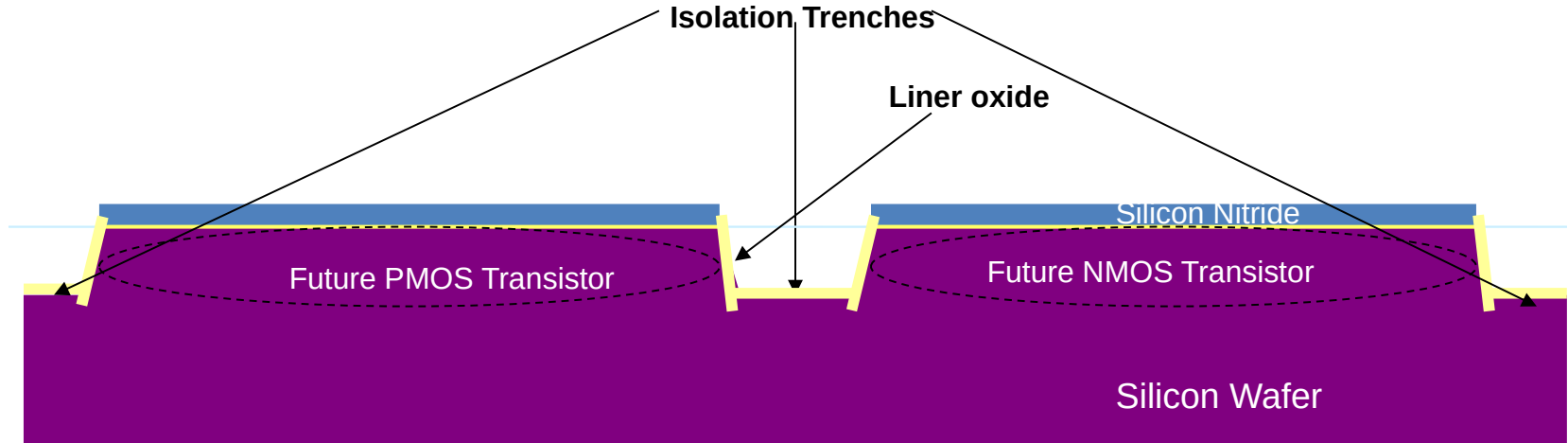
Remove Photoresist: An oxygen plasma is used to burn off the resist layer.



Shallow Trench Formation

Furnace STI – Trench Liner Oxide

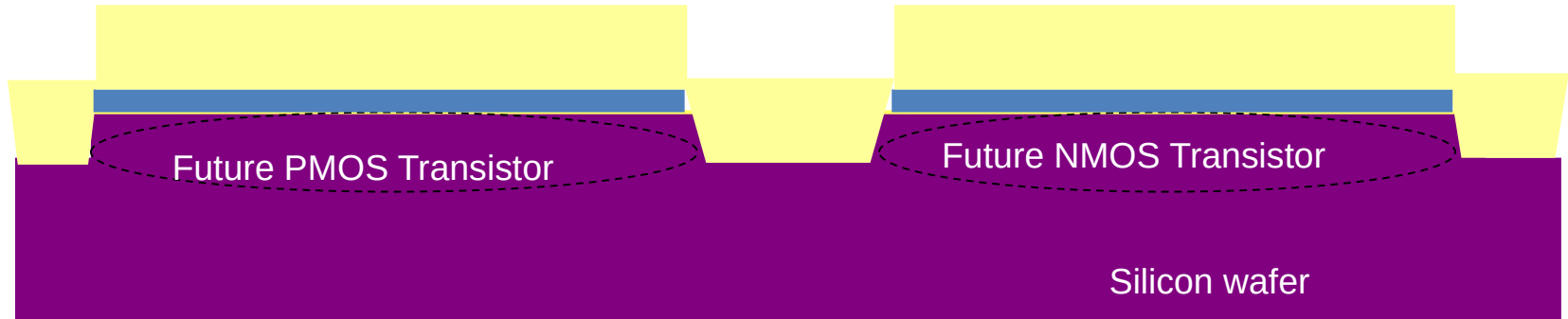
Liner Oxide Growth: Trench liner oxide is grown in atmospheric furnace.



Shallow Trench Formation

CVD – HDP Oxide Deposition

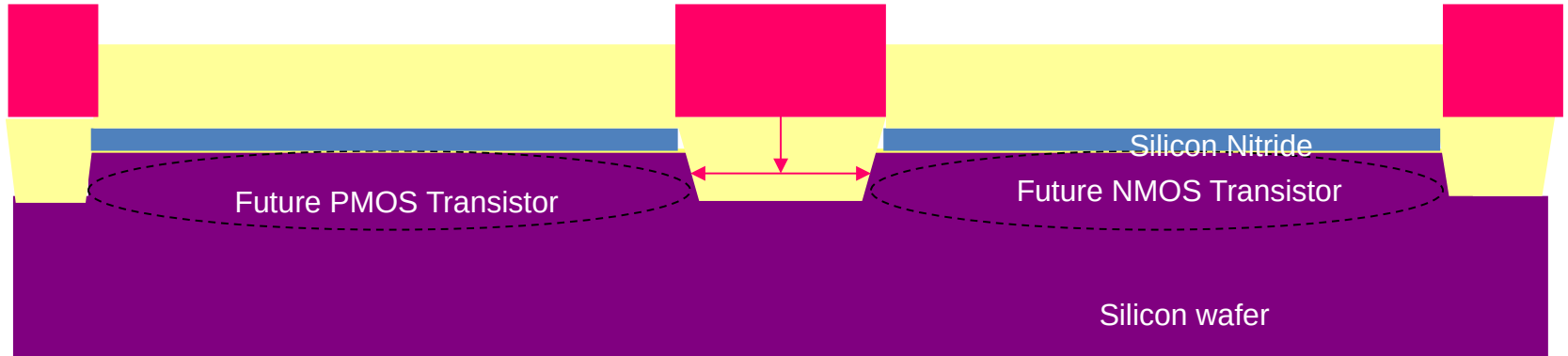
Fill Trenches with Oxide: A CVD oxide (HDP) layer is deposited to conformally fill the trenches. The oxide will prevent “cross-talk” between the transistors in the circuit.



RAA Mask

DUV Patterning

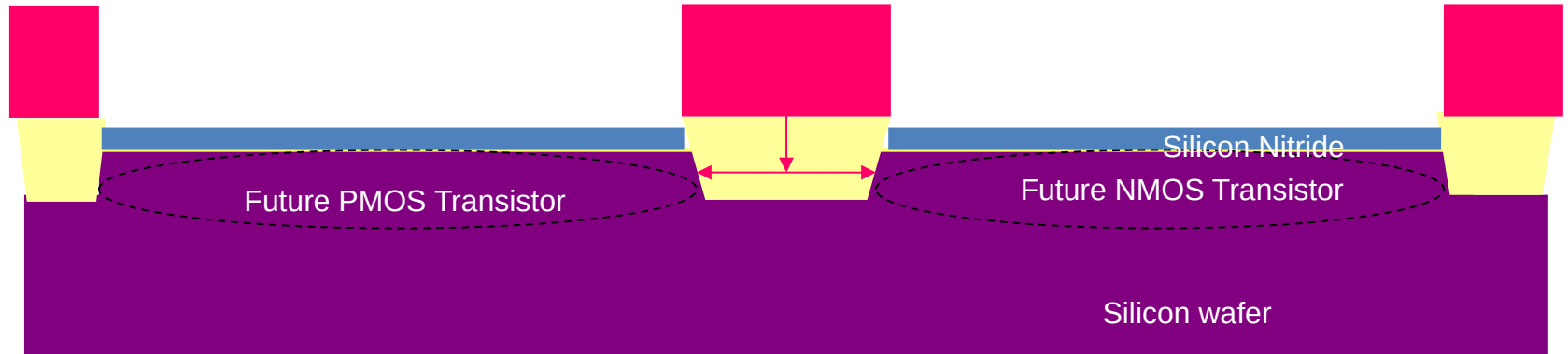
Reverse Active Area (RAA) Mask: Define areas to be oxide etch-back before CMP.



RAA ETCH

Oxide Etch

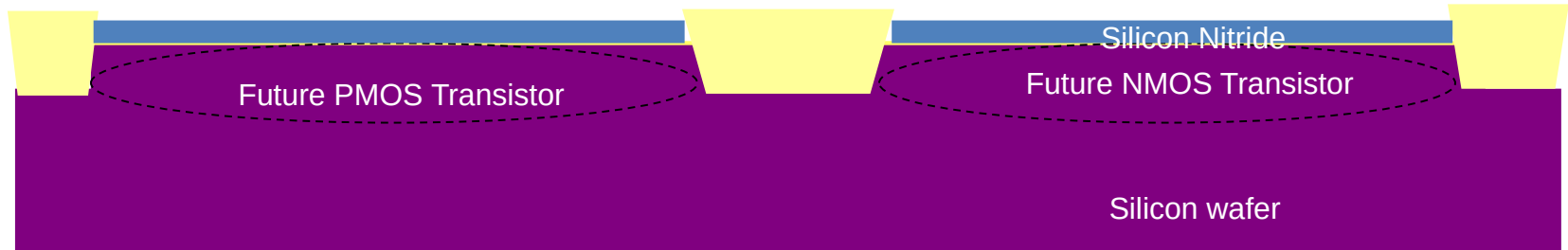
Reverse Active Area (RAA) Etch: Etch the HDP oxide at AA defined by the RAA mask by reactive ion etch (RIE) utilizing fluorine chemistry.



ASH PR

Dry Etch: Photo Resist Ashing

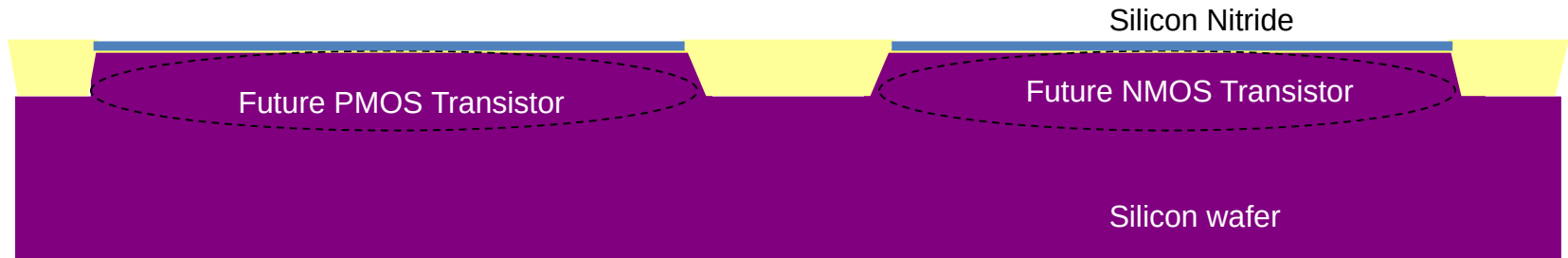
Remove Photoresist: An O_2/N_2 plasma is used to burn off the resist layer.



POLISH CMP

Oxide CMP

Oxide CMP: Extra deposited Oxide is removed and surface is planarized.



Shallow Trench Formation

Silicon Nitride Wet Etch

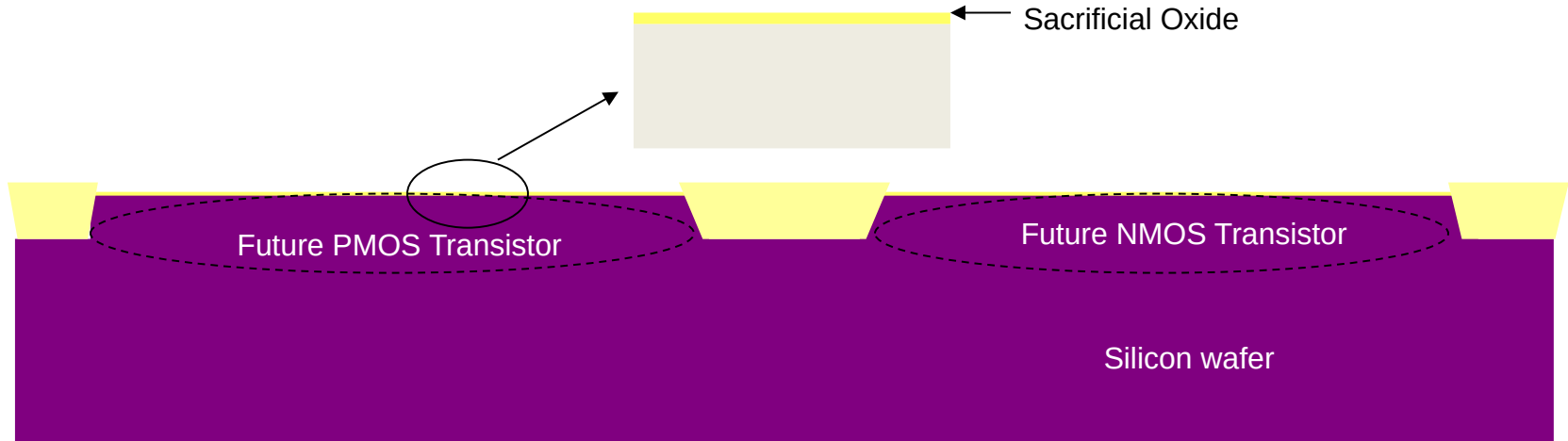
Remove Silicon Nitride & Oxide: A wet etch in hot phosphoric acid (H_3PO_4) is used to remove Silicon Nitride, completing formation of Shallow Trench Isolation (STI).



Shallow Trench Formation

Oxide Growth in Furnace

Grow Sacrificial Oxide: To protect the exposed Si surface of the active area from contamination during implant process.



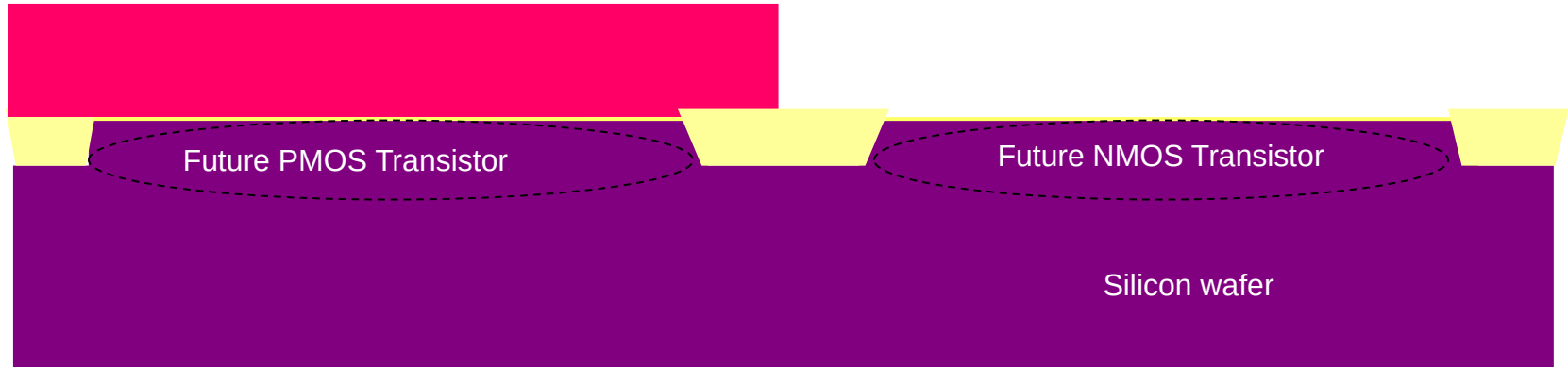
RETROGRADE WELLS

- **P-WELL**
 - P-Well mask
 - P-Well high energy implant
 - N-Channel channel punch-thru implant
 - N-Channel VT adjust implant
 - P-Well Comp Mask
 - P-Well Comp implant
- **N-WELL**
 - N-Well mask
 - N-Well high energy implant
 - P-Channel punch through implant
 - VTP-Implant
- **High energy implant damage anneal**

Well Formation

Patterning using MUV Track & MUV Stepper

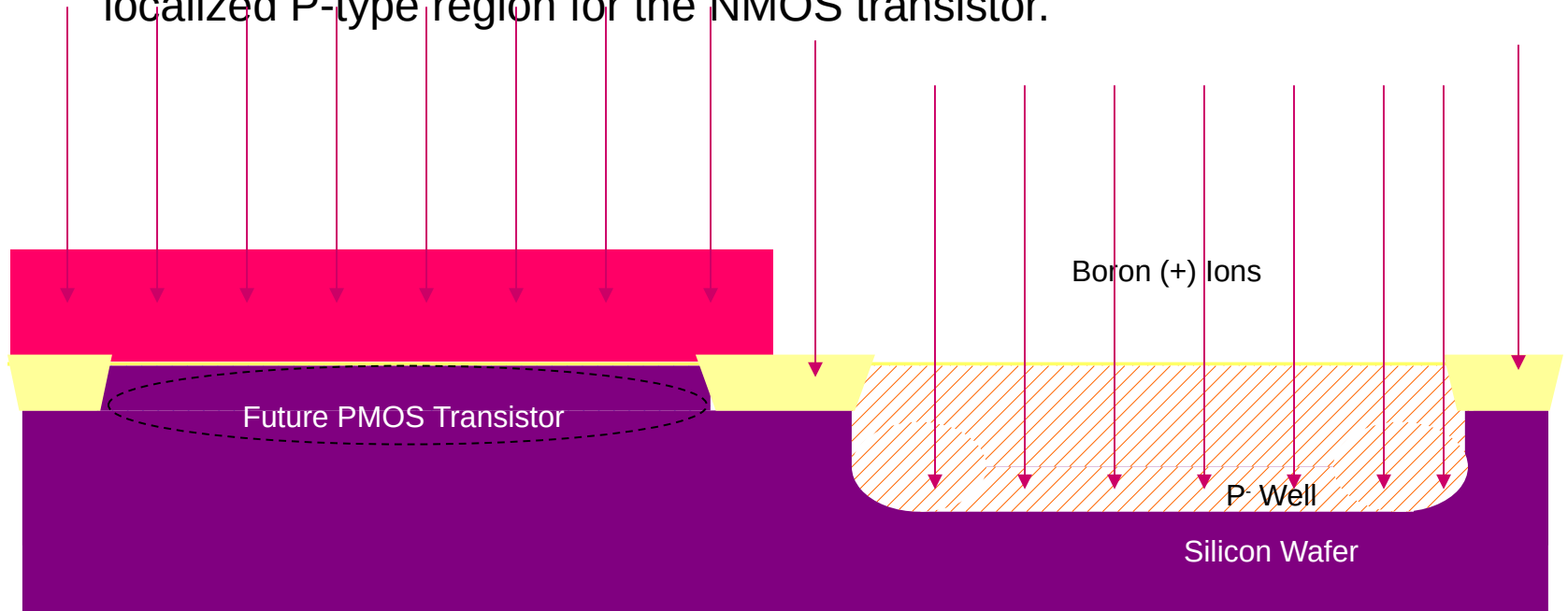
Pattern Photoresist for P-Well Formation: A non-critical masking layer, utilizing thicker resist to block the implant.



Well Formation

P Well

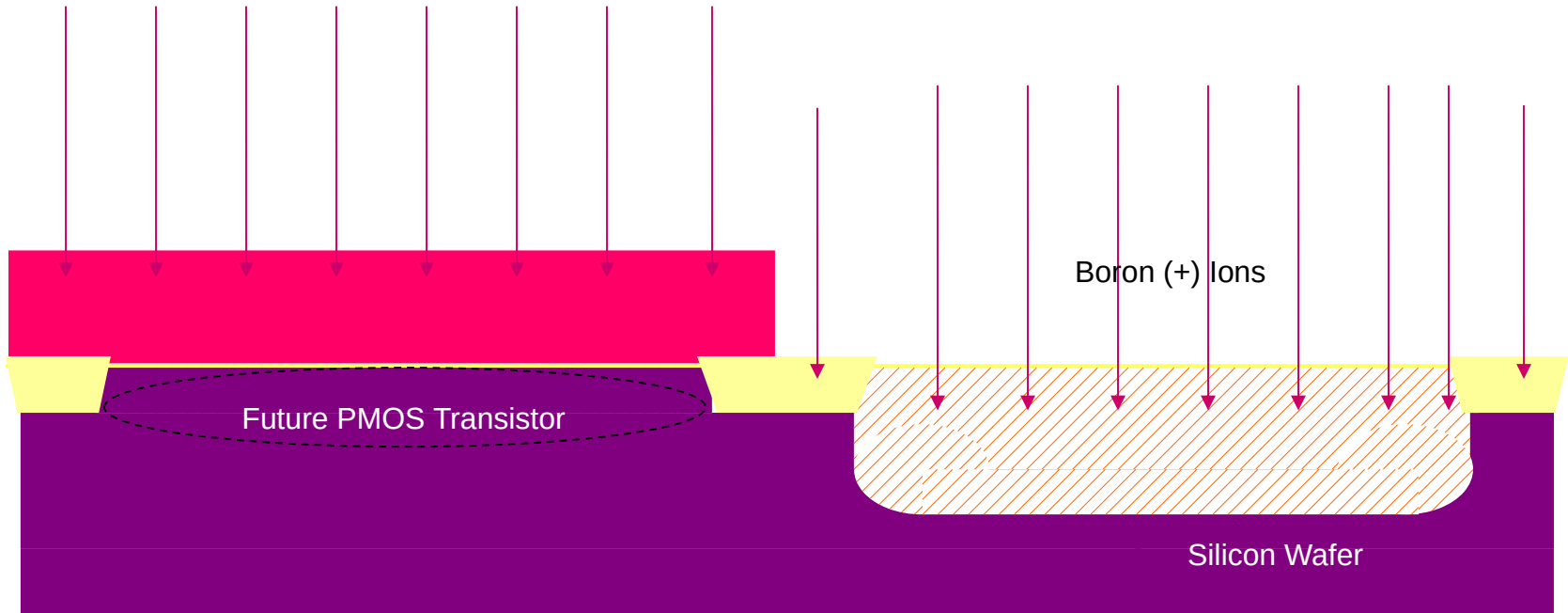
Implant P-Well: A deep (high-energy) implant of boron ions creates a localized P-type region for the NMOS transistor.



Well Formation

Punch Thru Implant

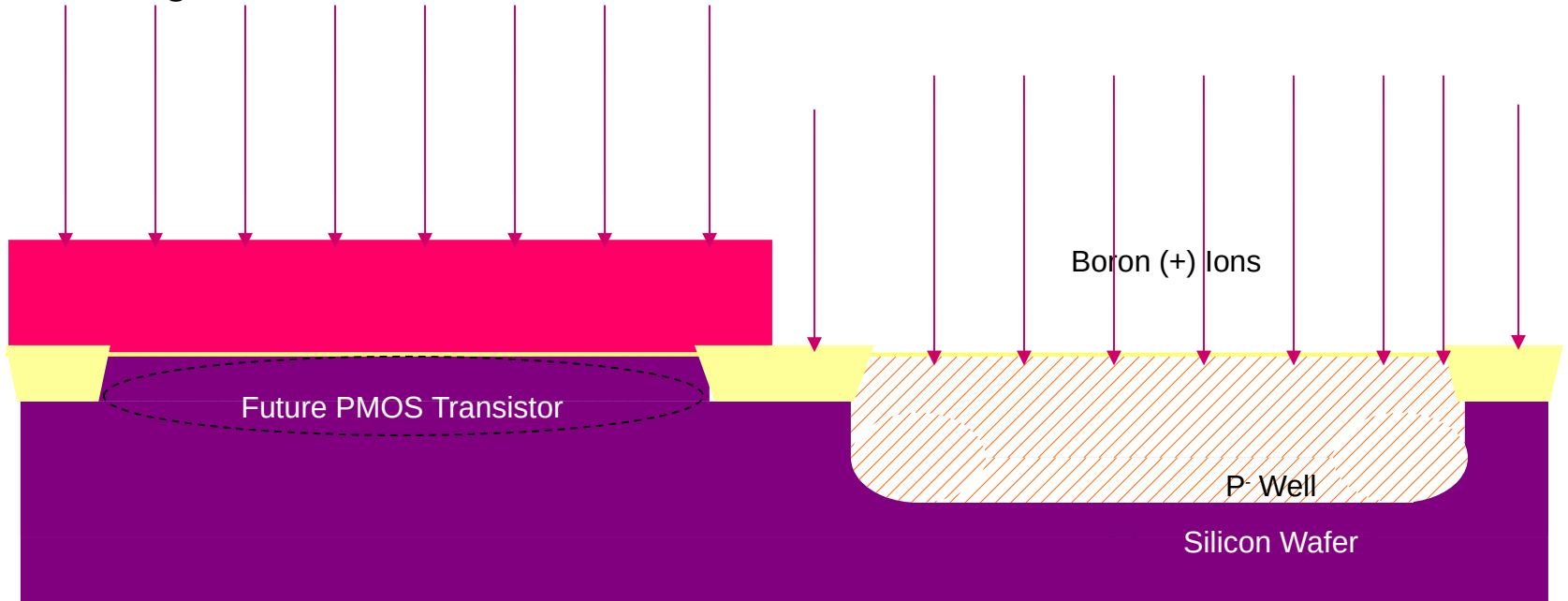
N-Channel punch through implant: This implant is required to reduce the successibility of punch through



Well Formation

Vt Adjustment Implant

N-Channel VT adjust implant: This implant is required to adjust the threshold voltage



Well Formation

PR Strip

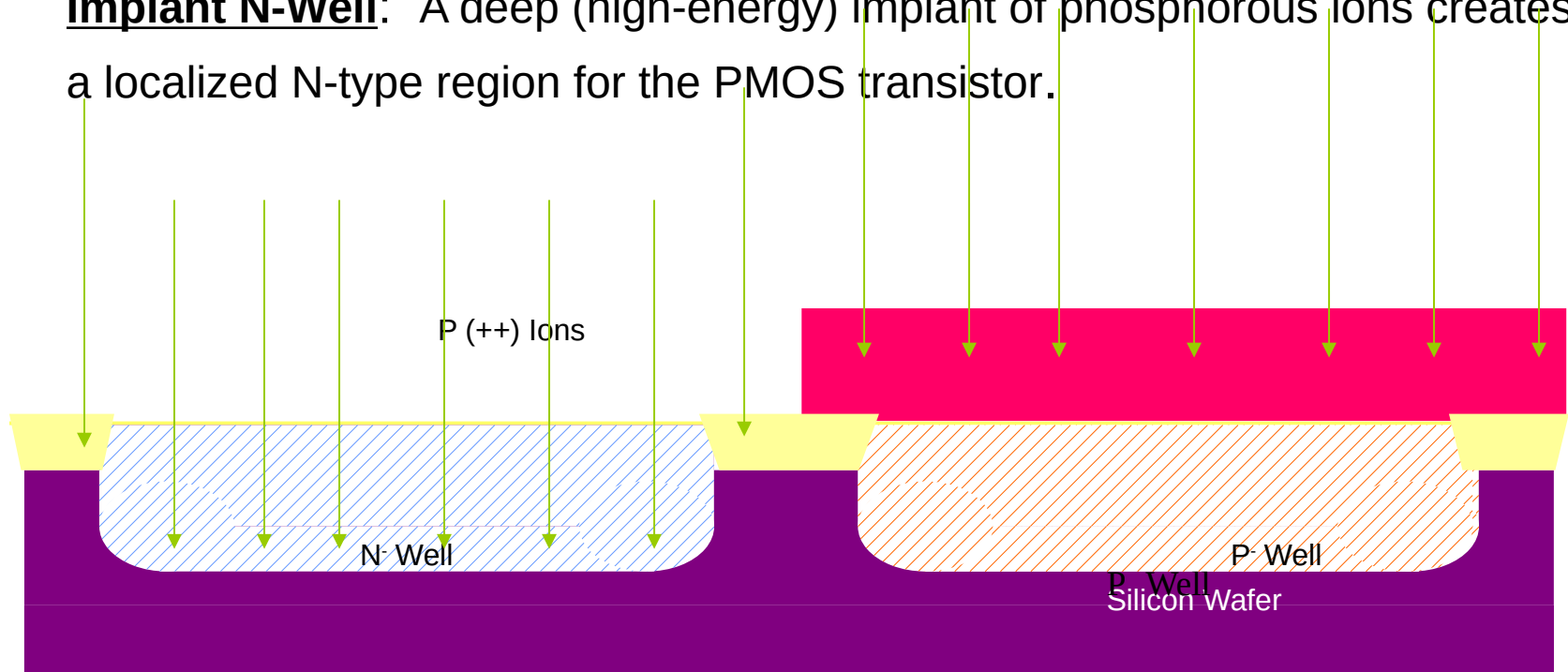
Strip N-Well Photoresist: Photoresist is stripped off using Piranha solution.



Well Formation

N-Well

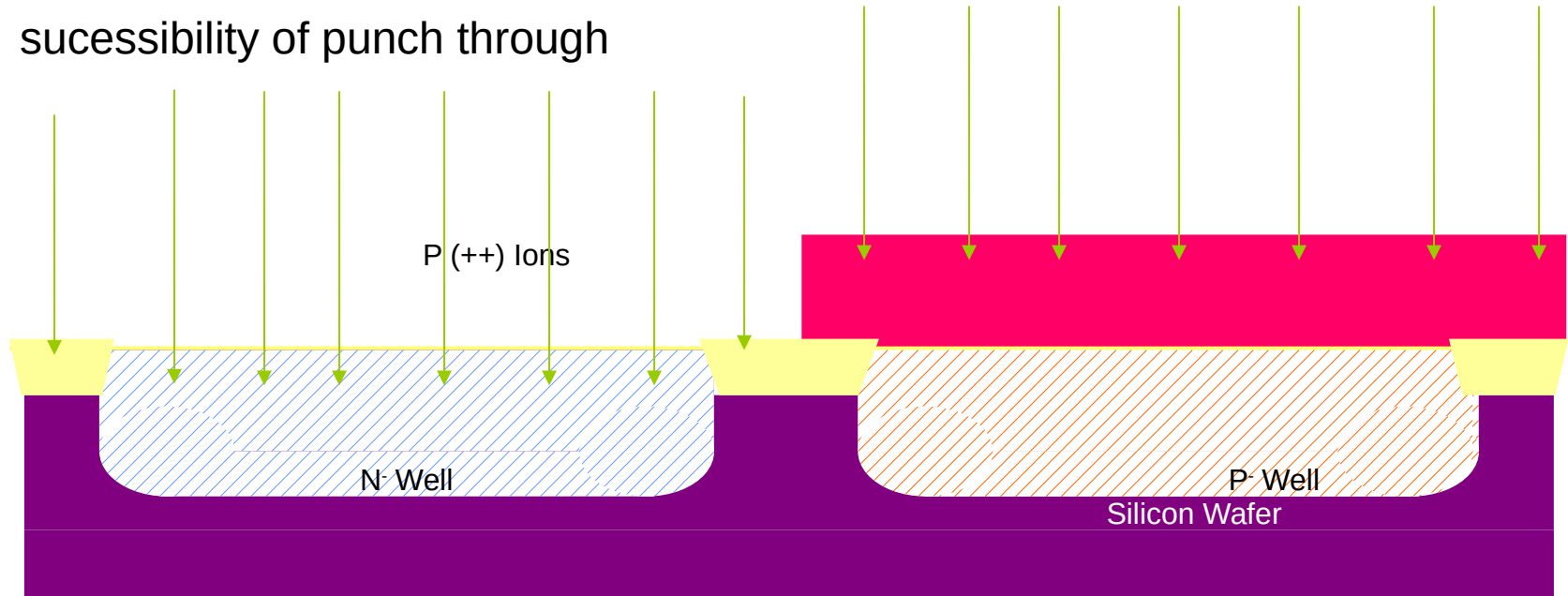
Implant N-Well: A deep (high-energy) implant of phosphorous ions creates a localized N-type region for the PMOS transistor.



Well Formation

Punch Thru Implant

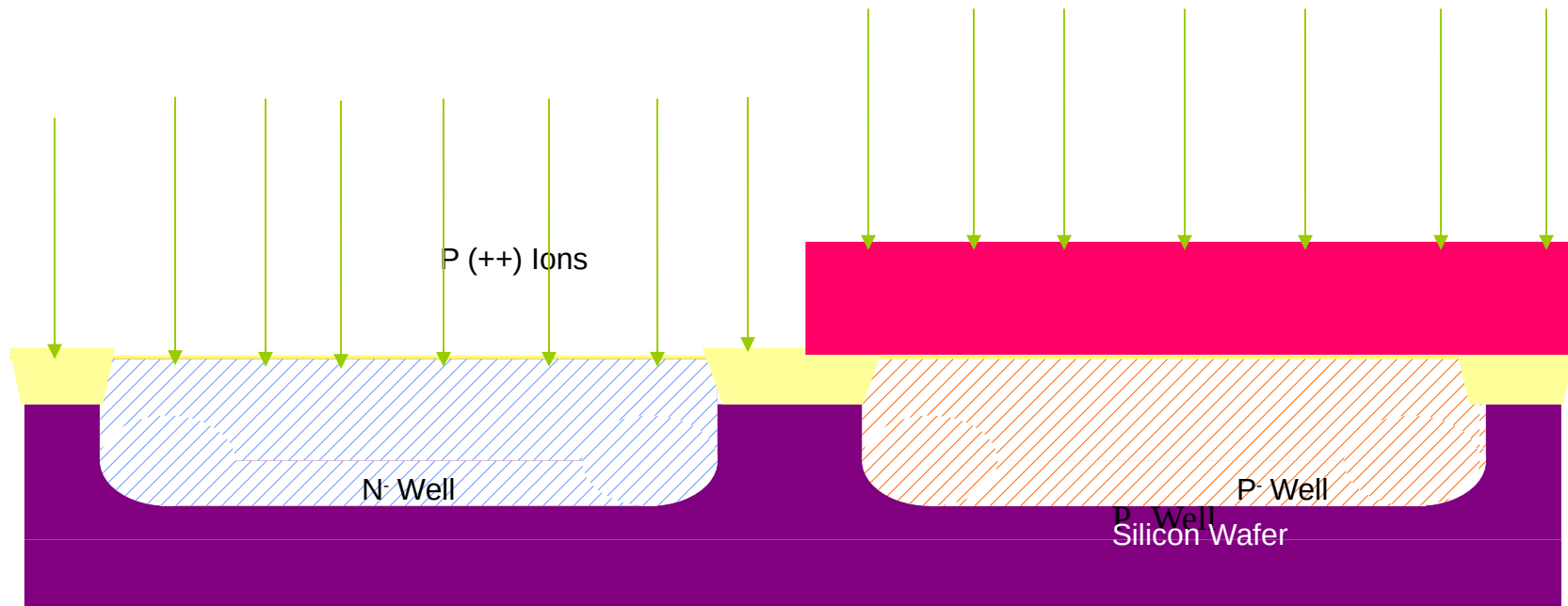
P-Channel punch through implant: This implant is required to reduce the successibility of punch through



Well Formation

Vt Adjustment Implant

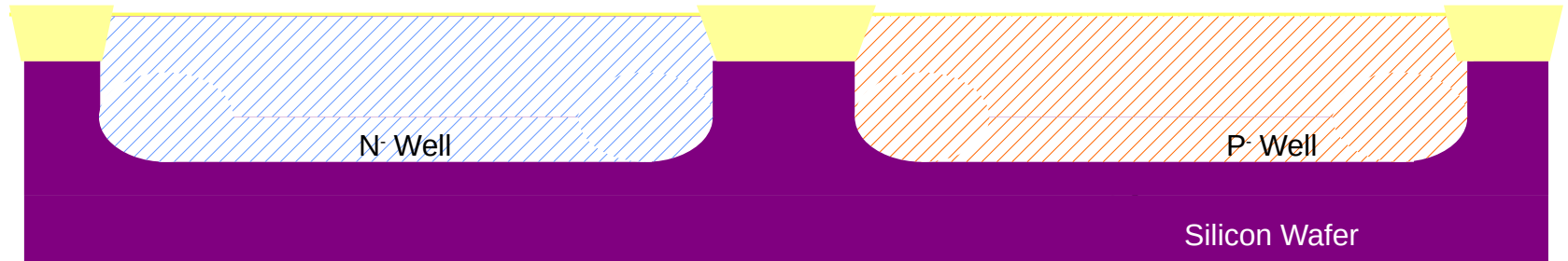
VTP-Implant: This implant is required to adjust the threshold voltage



Well Formation

PR Strip

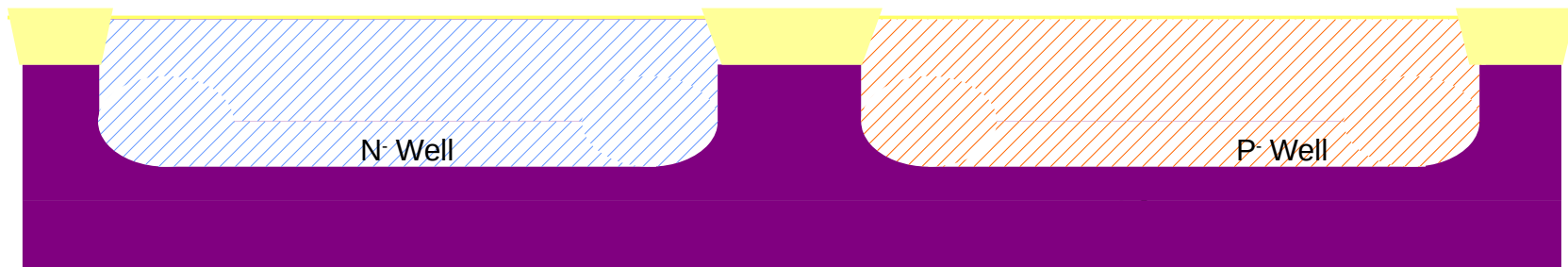
Strip P-Well Photoresist: Photoresist is stripped off using Piranha solution.



Well Formation

Rapid Thermal Anneal

Anneal Well Implants: This step repairs damage to the silicon surface caused by the implants and electrically activates the dopants. It also drives the dopants somewhat deeper, but Rapid Thermal Processing is used to minimize dopant spreading.



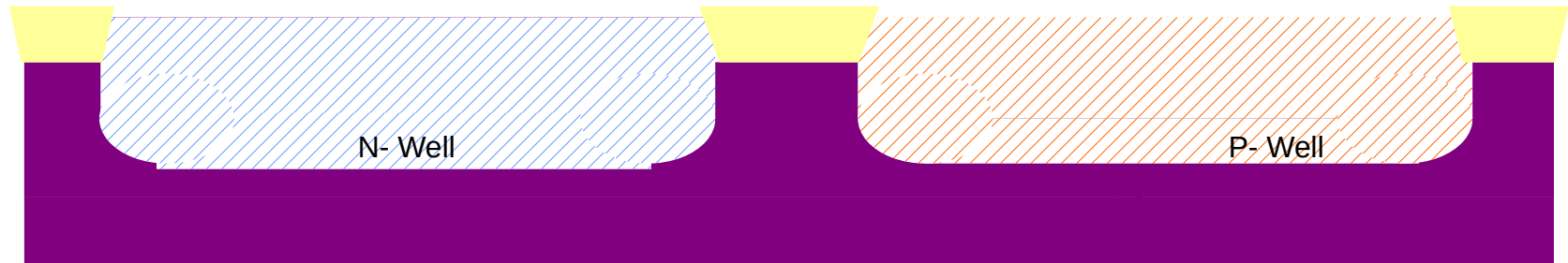
Gate Oxidation and Poly

- | | |
|--|--------------|
| – <u>Sac oxide removal & HV-GOX Growth</u> | Atm Furnace |
| – <u>HV-GOX mask</u> | MUV (i-line) |
| – <u>Oxide etch & LV-GOX growth</u> | Atm Furnace |
| – <u>Gate poly deposition</u> | LPCVD____ |
| – <u>Gate Mask</u> | DUV |
| – <u>Gate poly etch</u> | |

Gate Formation

Wet Pre Diffusion Clean

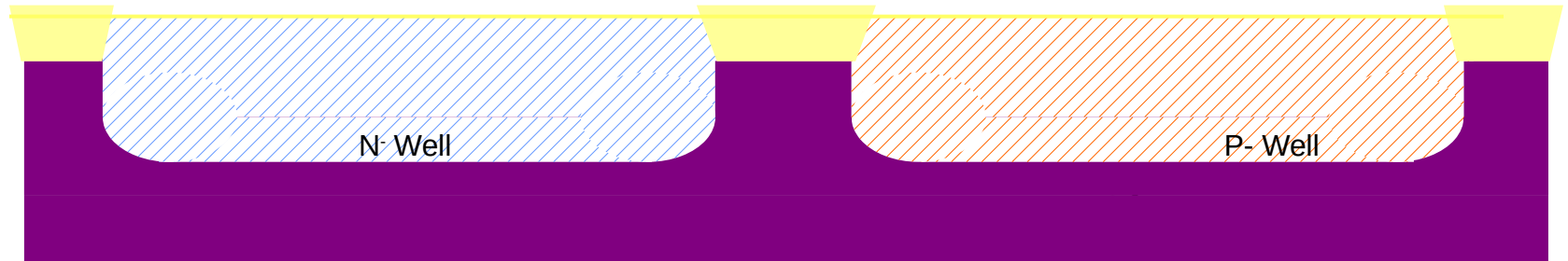
Remove Sacrificial Oxide: Sac ox is immediately removed in a wet HF solution, leaving behind a clean silicon surface.



Gate Formation

Furnace Gate Oxide

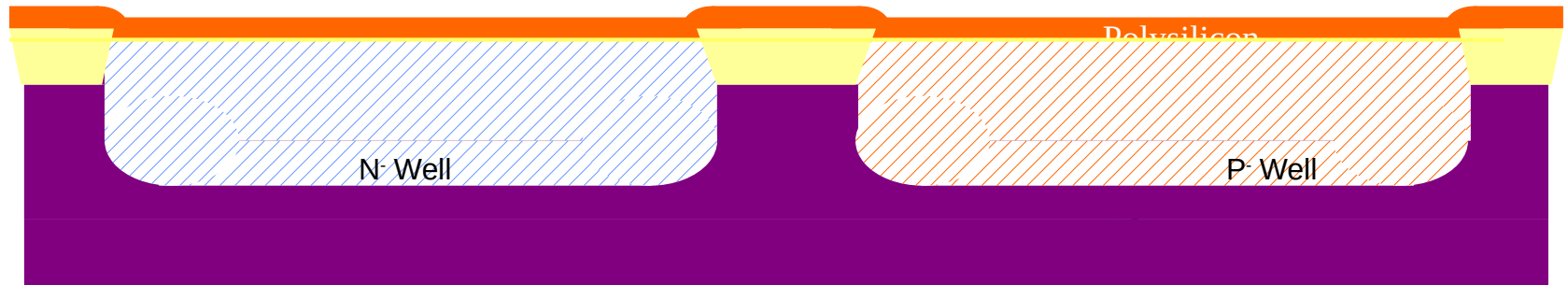
Grow Gate Oxide: *This is the most critical step in the process!* A very thin oxide layer is grown that will serve as the gate dielectric for both transistors. It must be extremely clean, and grown to a very precise thickness.



Gate Formation

Furnace Poly-Silicon Deposition

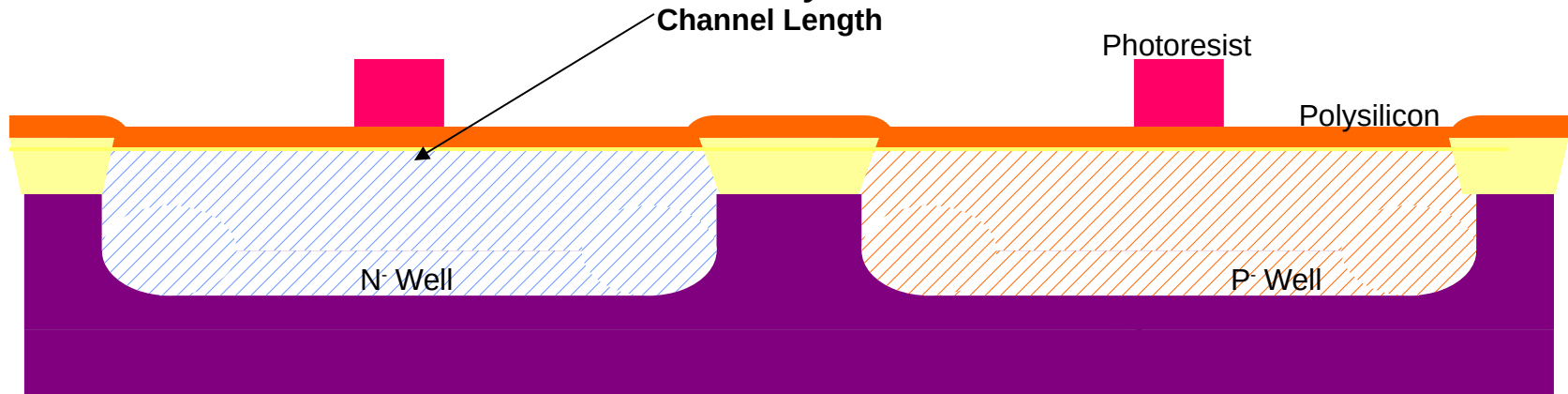
Deposit Polysilicon: Polycrystalline silicon is deposited using Chemical Vapor Deposition.



Gate Formation

DUV Track & DUV Scanner – Gate Patterning

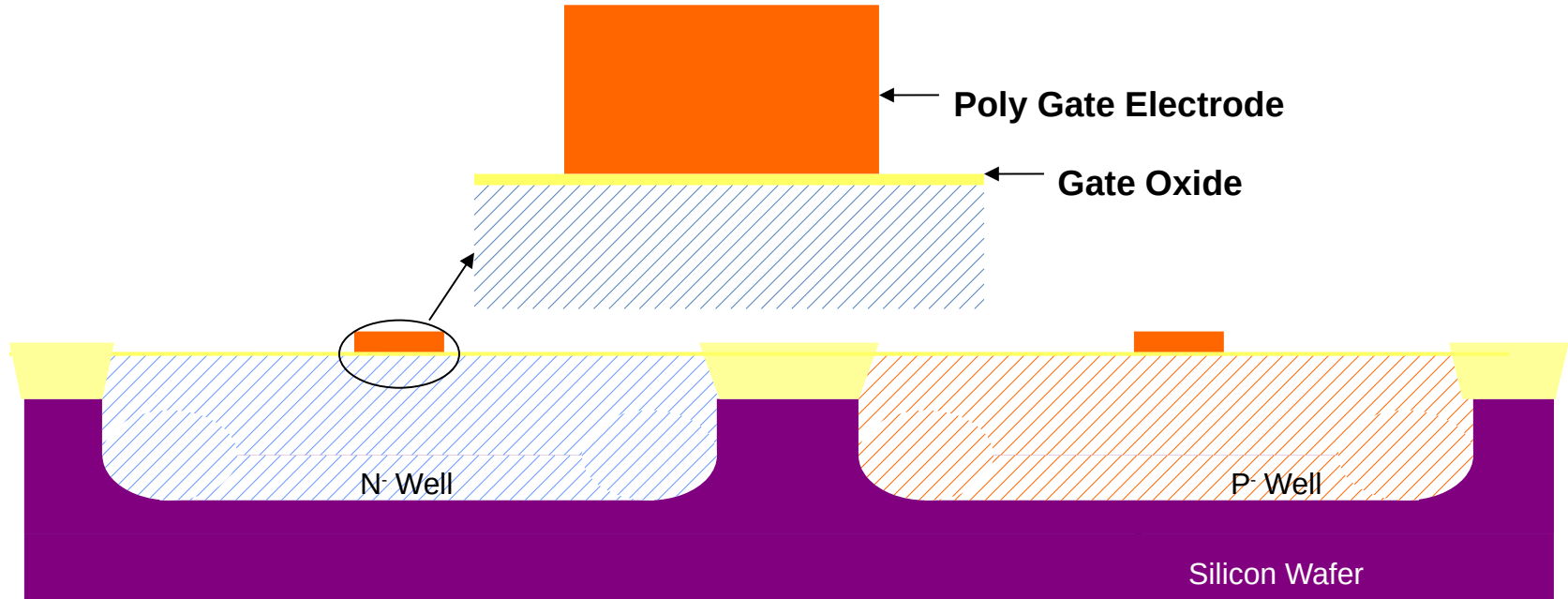
Pattern Photoresist to Define Gate Electrodes: *This is the most critical patterning step in the process!* Precise sizing of the poly gate length is a first-order determinant of transistor switching speed. The highest-technology patterning systems are used (i.e. DUV) along with thinner-than-normal photoresist due to the critical nature of the layer.



Gate Formation

Dry Etch – Gate Etch

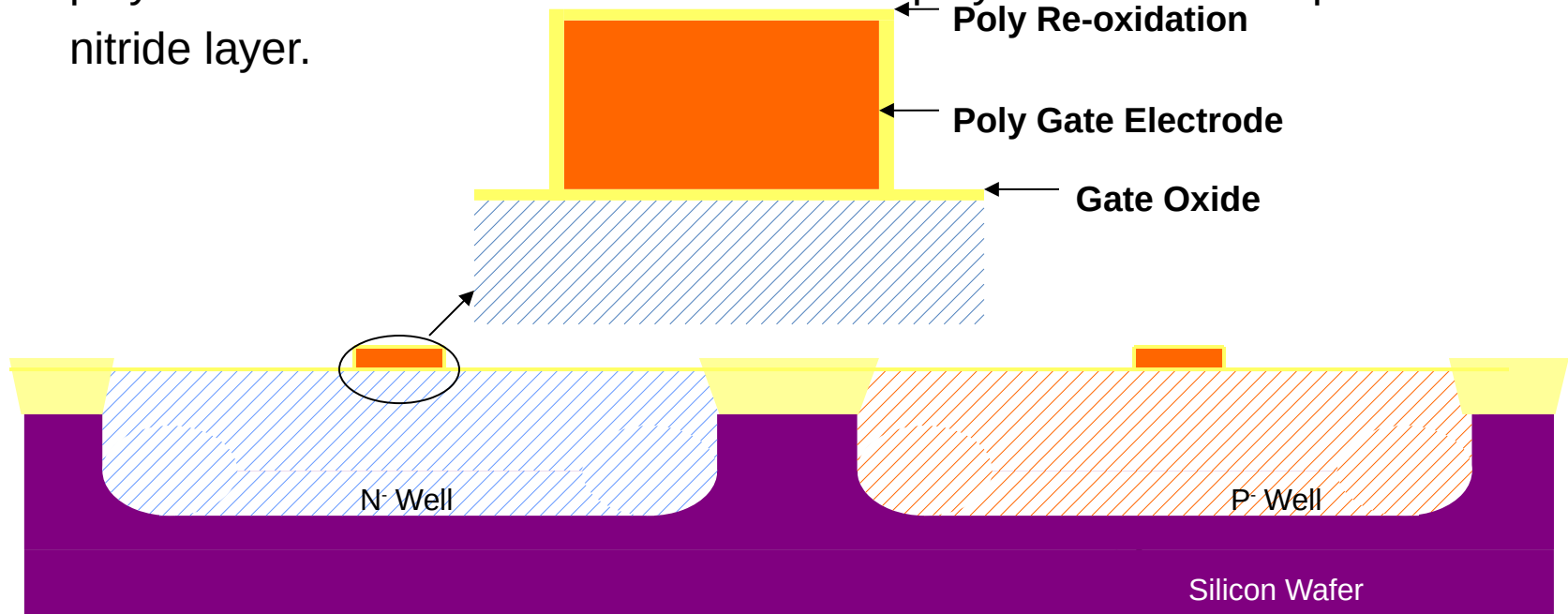
Etch Polysilicon and Strip Resist: Reactive Ion Etching using fluorine chemistry is used. This completes the formation of the “gate stack.”



Gate Formation

Furnace Oxide Growth

Oxidize Polysilicon: A thin layer of oxide is grown on top of the polysilicon to act as a buffer between the poly and the subsequent silicon nitride layer.



Shallow Source Drain (LDD) Extensions

- In the deep submicron MOSFET the source/drain contact structure consist of two region
 1. A shallow Source Drain Extension (SDE)
 2. A deeper contact source/drain region
- Role of SDE is to reduce short-channel effect
- The drain contact region is required to accommodate the thickness of the salicide contact layer and minimize junction -leakage

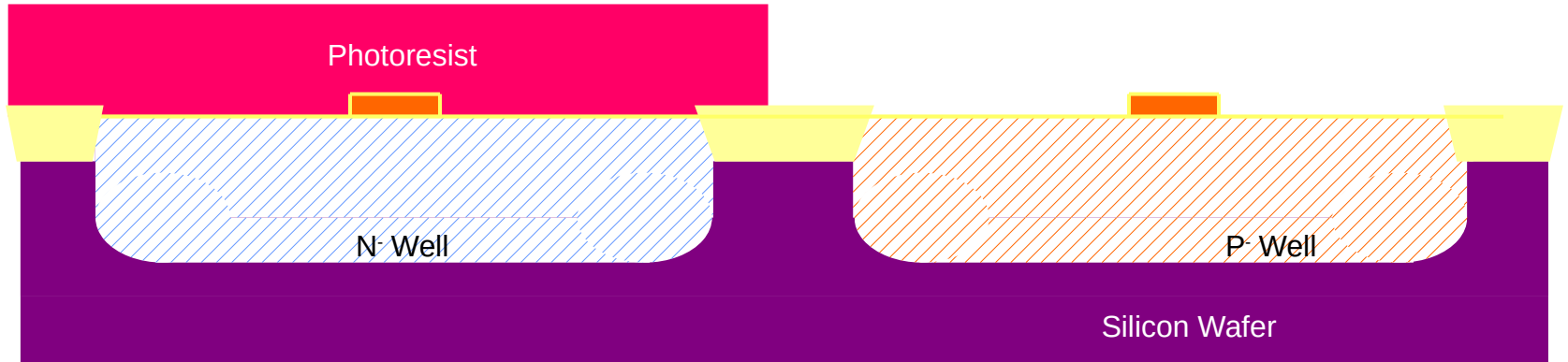
Shallow Source Drain (LDD) Extensions

- N-LDD mask MUV (i-line)
- N-S/D extension implant
- Pocket anti-punch through implant
- HV N-LDD mask MUV (i-line)
- HV-N- S/D extension implant
- HV P-LDD mask MUV (i-line)
- HV-P- S/D extension implant
- P-LDD mask MUV (i-line)
- P- S/D extension implant
- Pocket anti-punch through implant
- Damage anneal

Source/Drain LDD Formation

MUV Track & MUV Stepper - Patterning

Pattern Photoresist for NMOS Transistor SDE

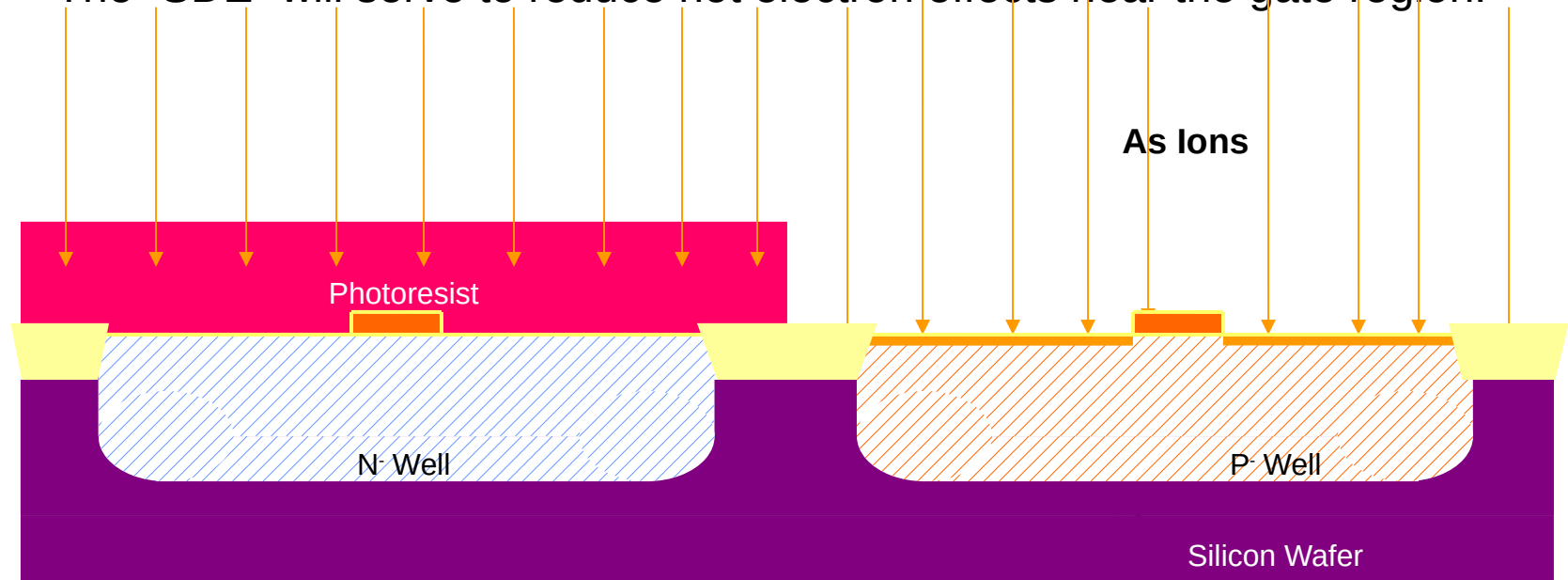


Source/Drain Extension Formation

SDE Implant

NMOS Transistor SDE Implant: A very shallow (low energy) implant of arsenic ions begins the formation of the NMOS transistor source and drain.

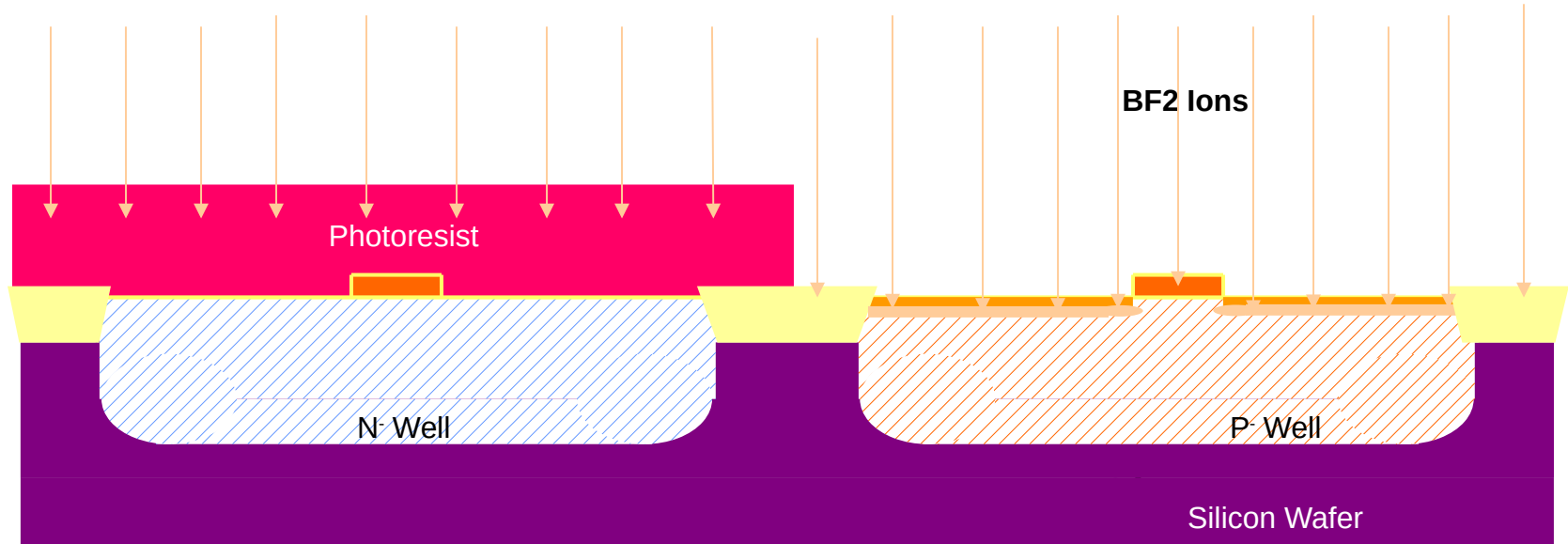
The “SDE” will serve to reduce hot electron effects near the gate region.



Source/Drain Extension Formation

HALO/ PCKET Implant

Pocket anti-punch through implant: Suppresses punchthrough current.

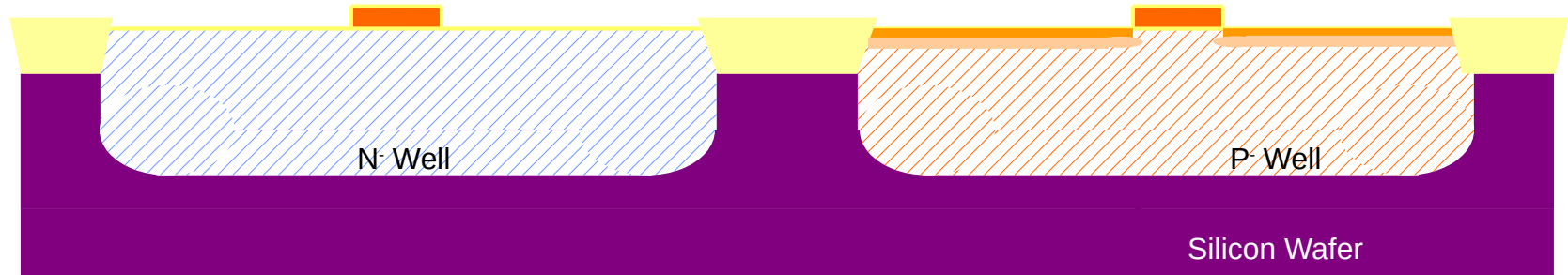


Source/Drain Extension Formation

Ash & Wet PR Strip

PR Ashing : An O_2/N_2 plasma is used to burn off the resist layer.

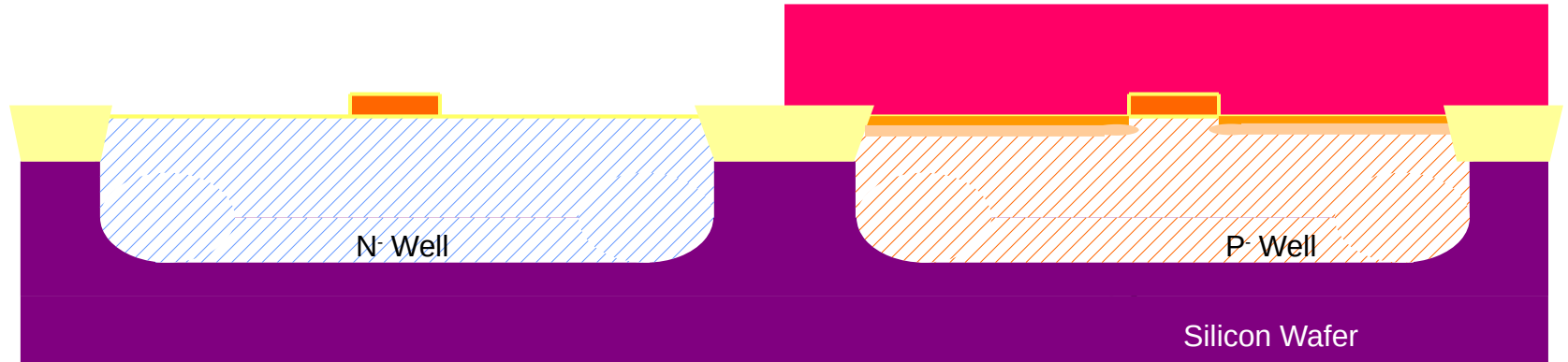
Strip Photoresist: Photoresist is stripped off using Piranha solution.



Source/Drain Extension Formation

MUV Track & MUV Stepper – Patterning

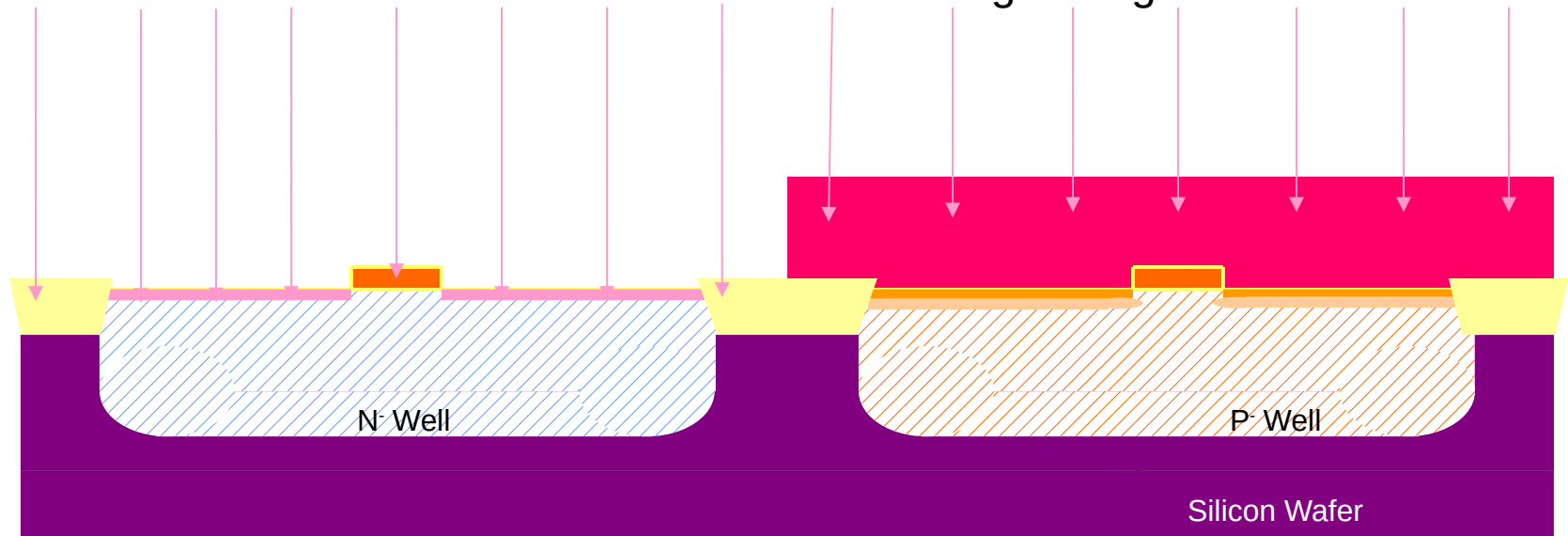
Pattern Photoresist for PMOS Transistor SDE Implant



Source/Drain Extension Formation

SDE Implant

PMOS Transistor SDE Implant: A very shallow (low energy) implant of BF₂ ions begins the formation of the PMOS transistor source and drain. The “tip” will serve to reduce hot electron effects near the gate region.



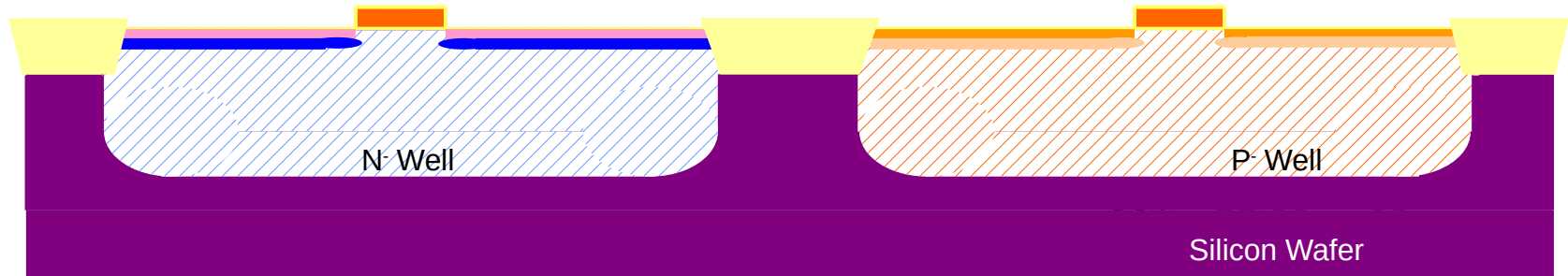
Source/Drain Extension Formation

Ash, Wet PR Strip, and Rapid Thermal Anneal

PR Ashing : An O_2/N_2 plasma is used to burn off the resist layer.

Strip Photoresist: Photoresist is stripped off using Piranha solution.

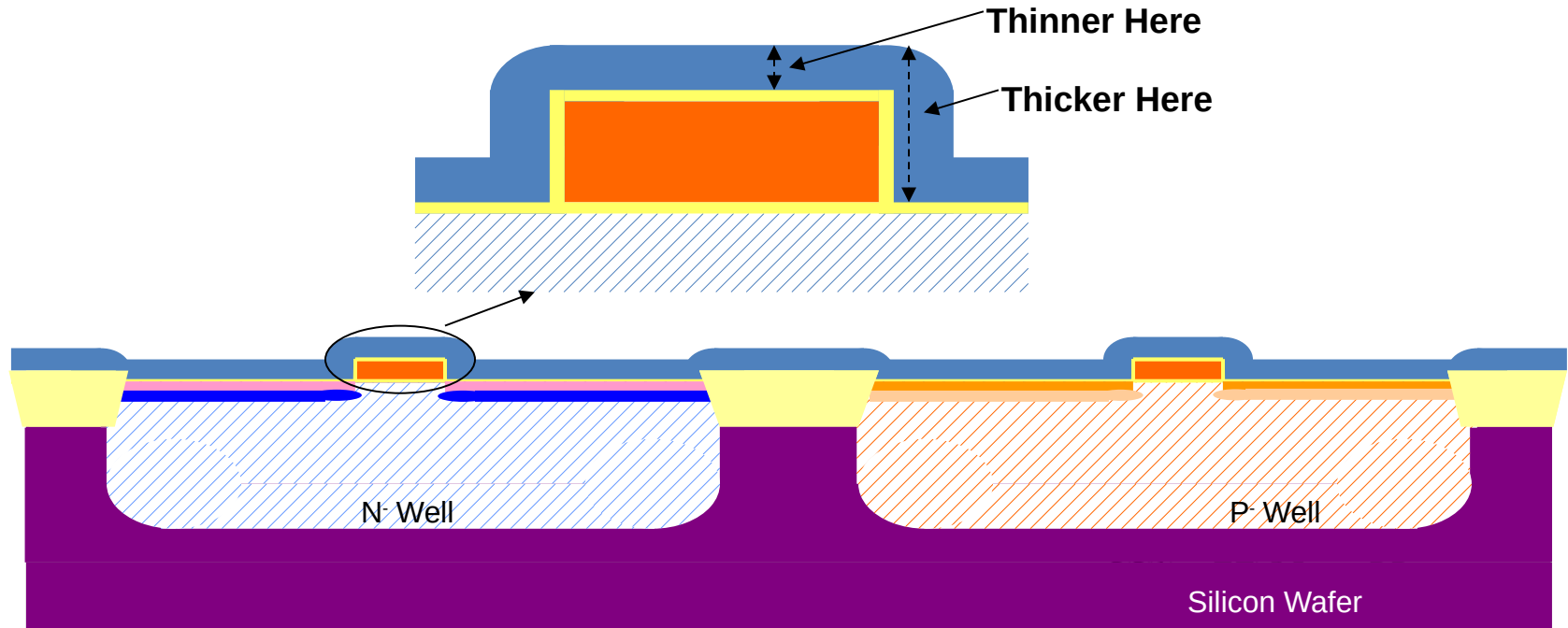
Damage Anneal Strip Photoresist: Heal the silicon surface from damages created during the implantations and activates implants.



Spacer Formation

Furnace TEOS Oxide & Furnace Si Nitride

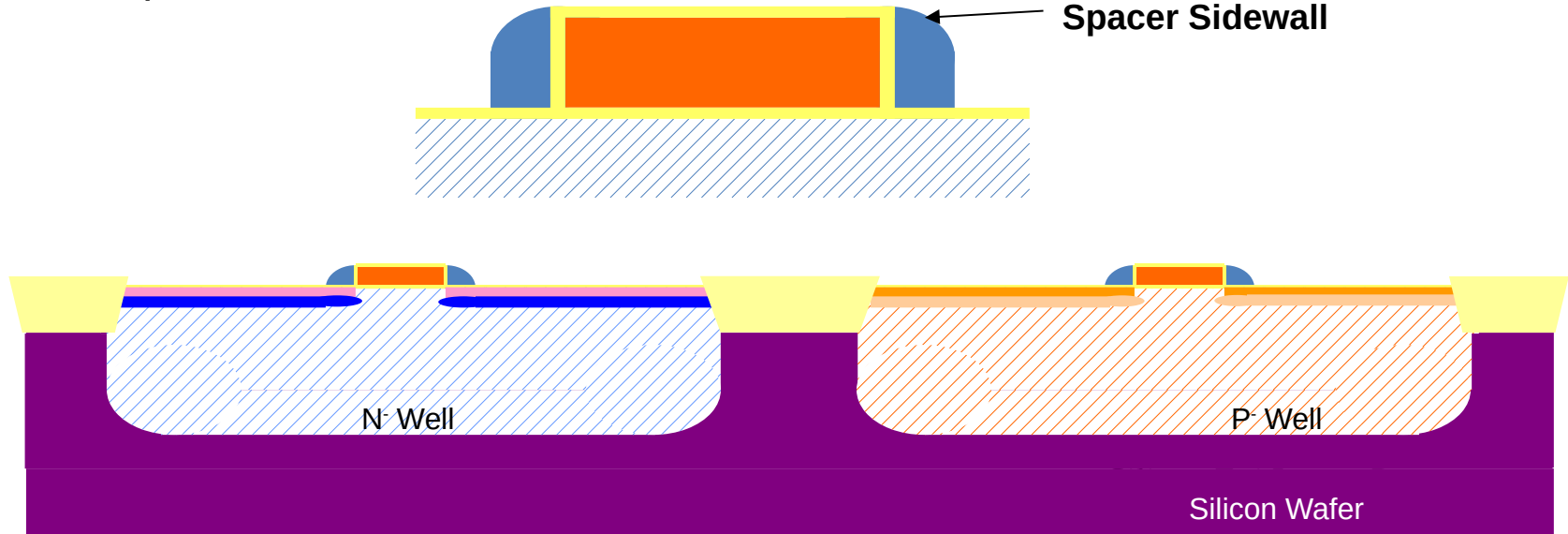
Deposit oxide & Silicon Nitride Layer: Using Chemical Vapor Deposition.



Spacer Formation

Dry Etch

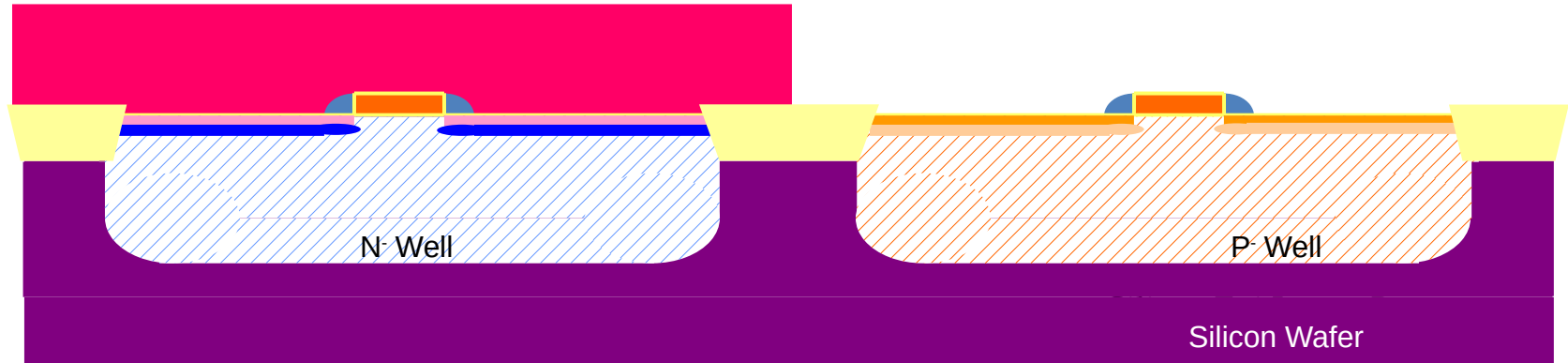
Etch Nitride & Oxide to Form Spacer Sidewalls: Using a carefully controlled RIE etch, the thin nitride & oxide is removed from the horizontal surfaces, but the sidewalls remain. These sidewalls will precisely position the implants that form the transistor sources and drains.



Source/Drain Formation

MUV Track & MUV Stepper

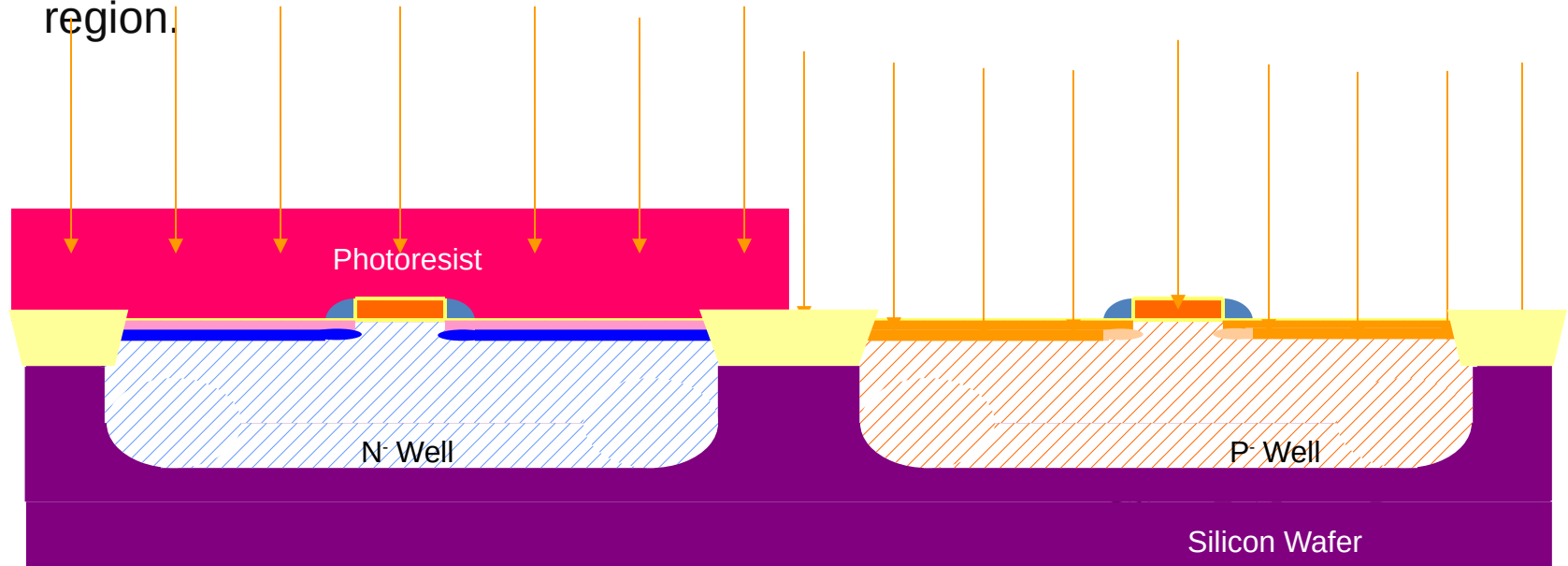
Pattern Photoresist for NMOS Transistor Source/Drain Implant:



Source/Drain Formation

Source/ Drain Implant

NMOS Transistor Source/Drain Implant: A shallow and high-dose implant of arsenic ions completes the formation of the heavily-doped NMOS transistor source and drain. The spacer shadows the implant near the gate region.

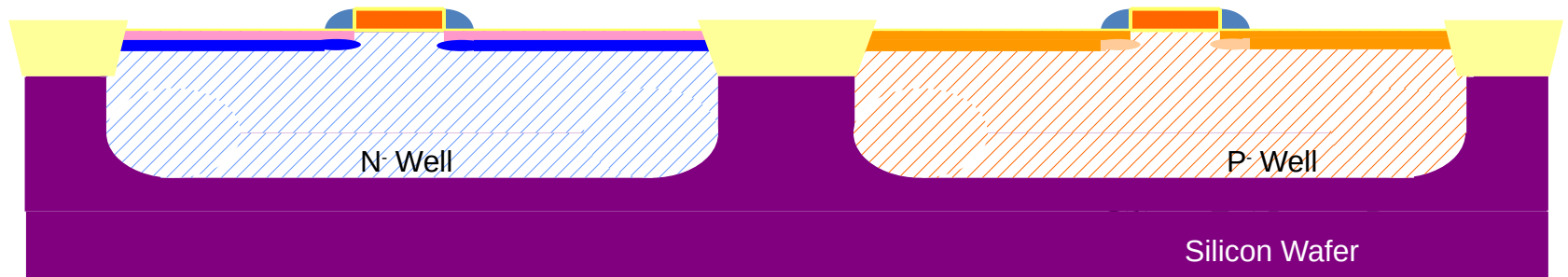


Source/Drain Formation

Asher & Wet PR Strip

PR Ashing : An O_2/N_2 plasma is used to burn off the resist layer.

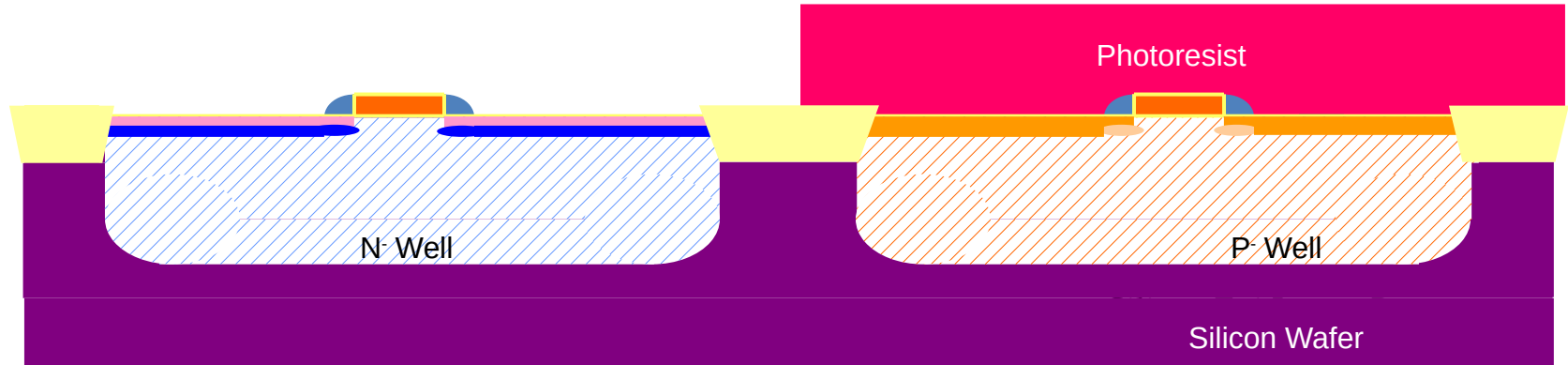
Strip Photoresist: Photoresist is stripped off using Piranha solution.



Source/Drain Formation

MUV Track & MUV Stepper – Patterning

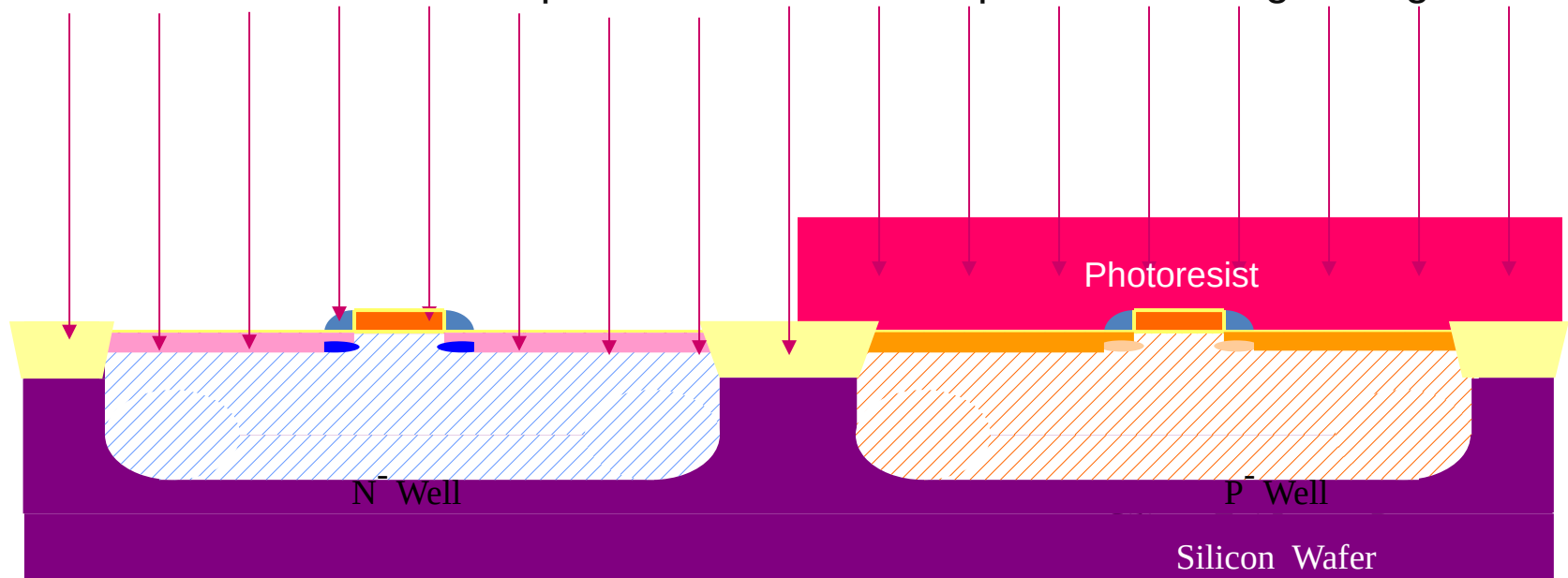
Pattern Photoresist for PMOS Transistor Source/Drain Implant



Source/Drain Formation

Source/Drain Implant

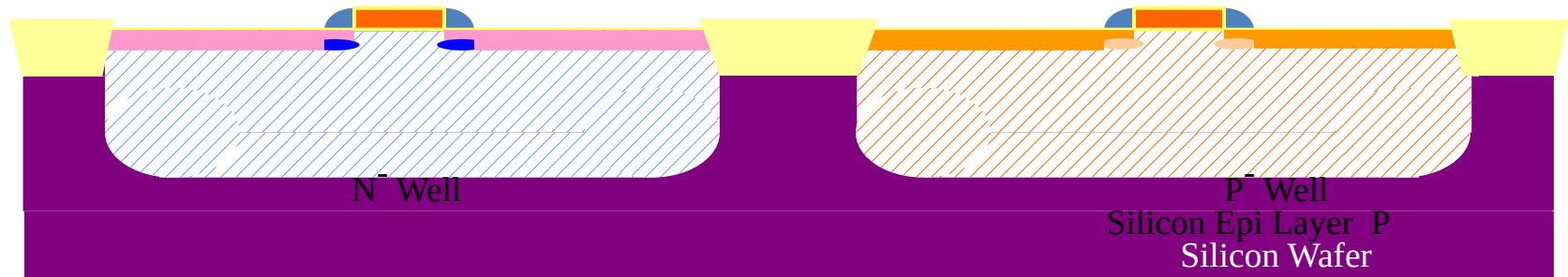
PMOS Transistor Source/Drain Implant: A shallow and high-dose implant of B ions completes the formation of the heavily-doped PMOS transistor source and drain. The spacer shadows the implant near the gate region.



Source/Drain Formation

Asher & Wet PR Strip & Rapid Thermal Anneal

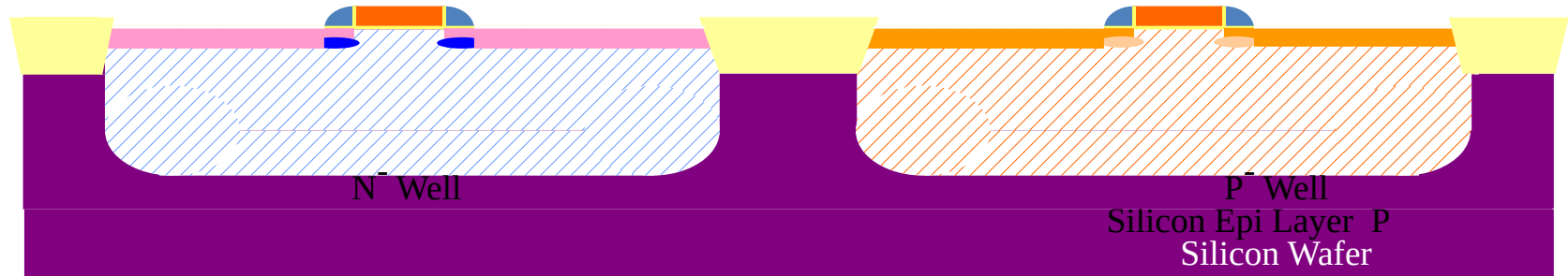
Strip Photoresist and Anneal Implants: Use Rapid Thermal Annealing to virtually eliminate dopant migration in the shallow source and drains. The electronic devices are now completely formed. All that remains is to connect them together.



Salicide Formation

Wet oxide strip

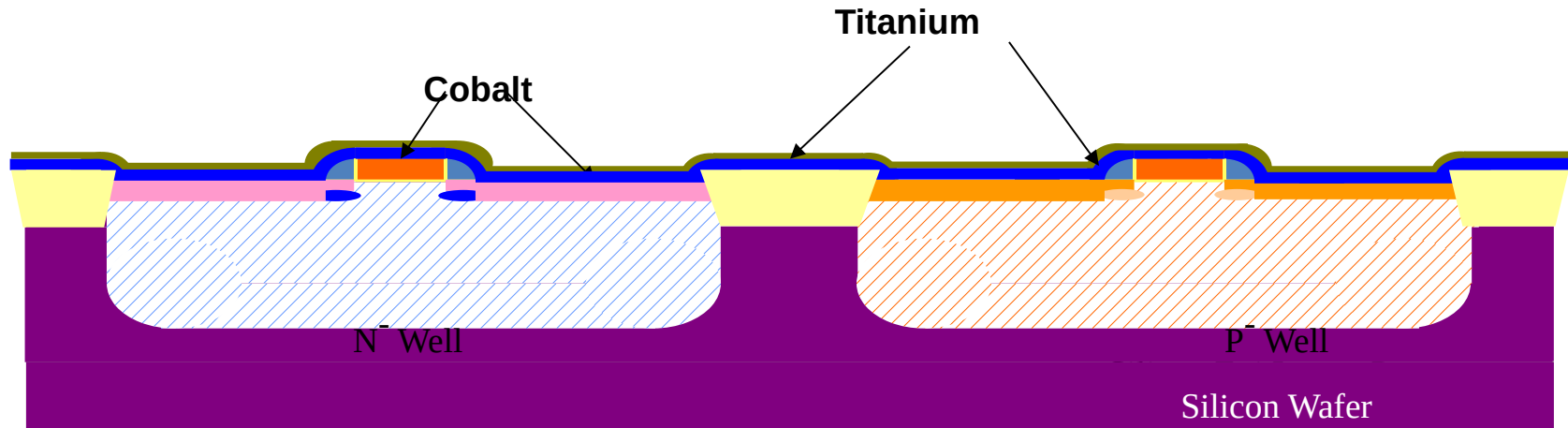
Strip Surface Oxides: A quick dip in HF to expose bare silicon in the source, gate, and drain areas.



Salicide Formation

Sputter

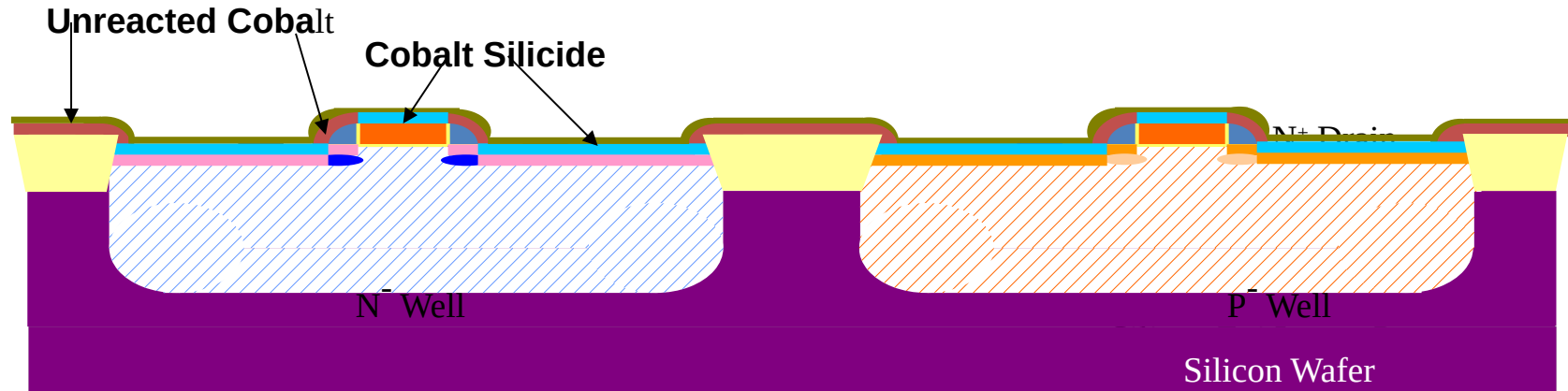
Deposit Cobalt & Titanium : Use a sputter tool to deposit a thin layer of cobalt across the entire wafer surface. After cobalt deposition titanium is deposited



Salicide Formation

Rapid Thermal Anneal

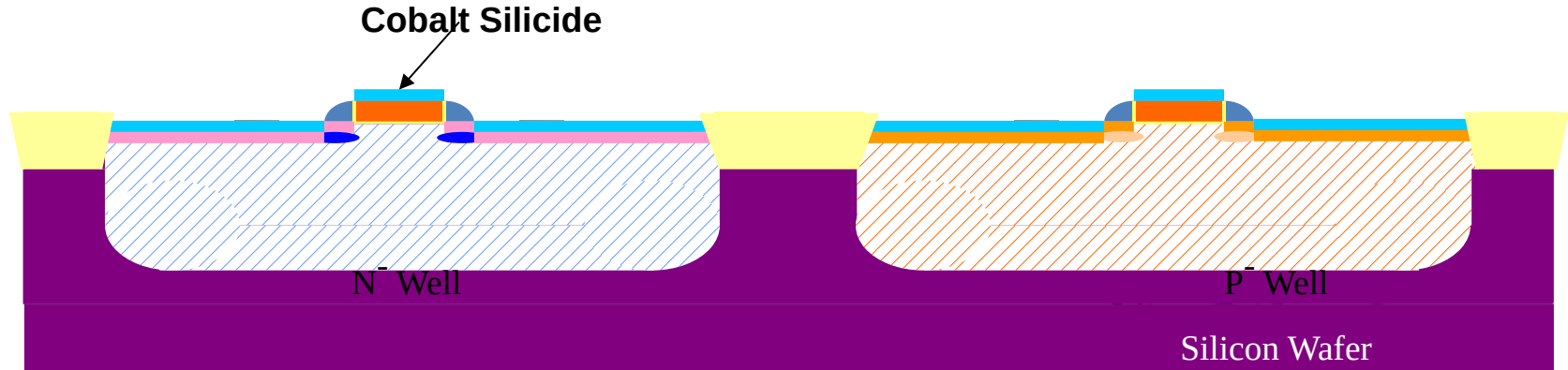
Cobalt Silicide Formation: Rapid Thermal Processing causes the cobalt to react with silicon, forming cobalt silicide, where the two are in contact. In other areas, the cobalt is unchanged. This process perfectly aligns the silicide to the exposed silicon, and is called Self-Aligned Silicide, or Salicide.



Salicide Formation

Wet Etch & Rapid Thermal Anneal

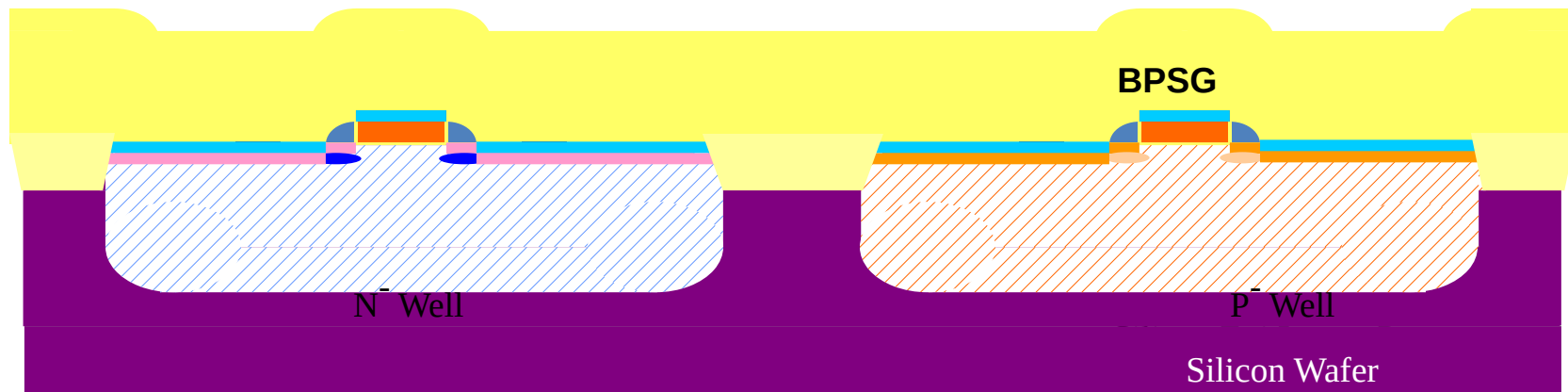
Titanium Nitride & Cobalt Etch: Titanium nitride & unreacted cobalt is removed using wet strip Piranha solution. The cobalt silicide remains. Then a high temp RTP is given to get stable phase of CoSi_2 . It provides an ohmic contact between silicon and metal.



1st Interconnect Layer

CVD Nitride, APCVD BPSG, and CVD TEOS

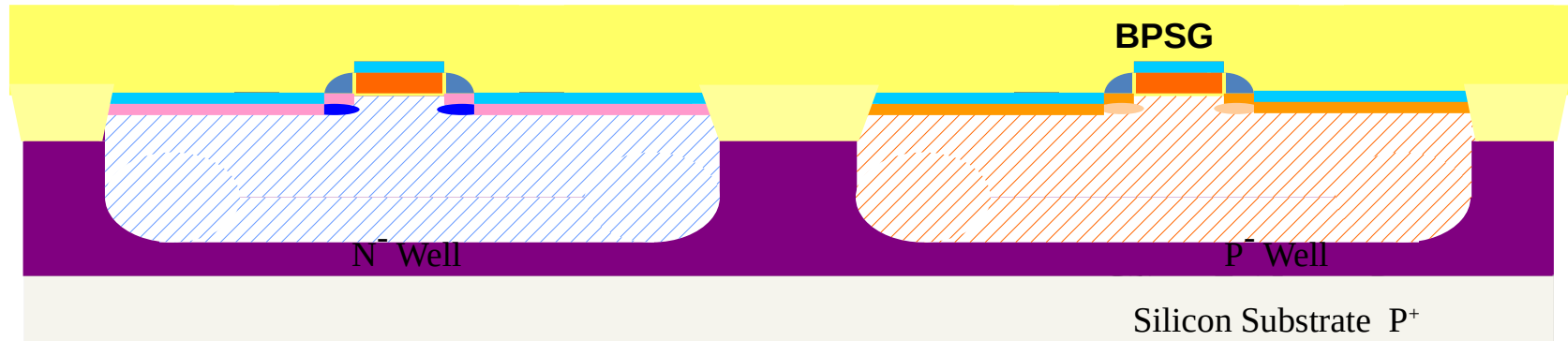
Deposit BPSG & TEOS Oxide: Silicon dioxide doped with small amounts of boron and phosphorous to enable film reflow and to getter contaminants is deposited using Chemical Vapor Deposition. After this TEOS oxide deposited. This layer will electrically insulate the devices from the 1st metal layer.



1st Interconnect Layer

Oxide CMP

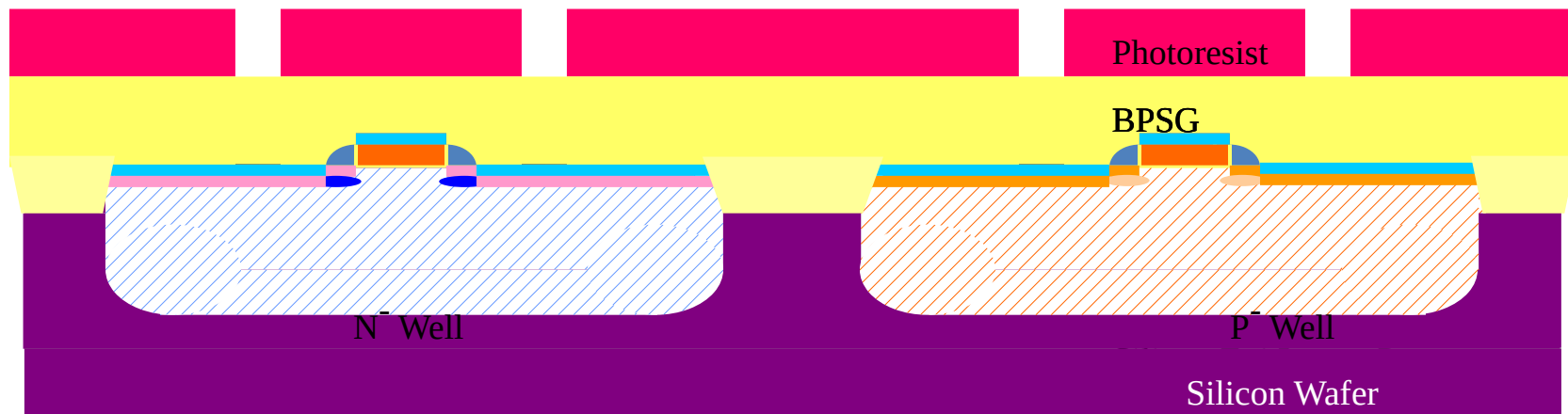
Polish BPSG: Use Chemical Mechanical Polishing to achieve a flat surface on the Oxide layer. If not removed, the bumps on the surface from the underlying topography would cause a problem for the subsequent photolithography steps and degrade metal step coverage.



1st Interconnect Layer

DUV Track & DUV Scanner – Patterning

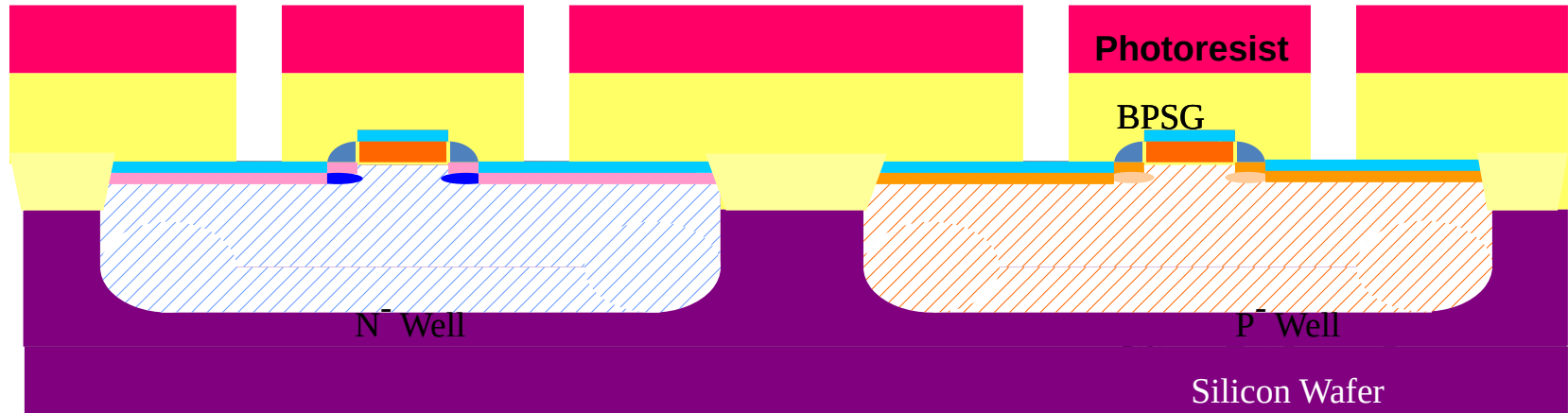
Pattern Photoresist to Define Contacts: Contacts are openings in the oxide layer enabling electrical access to the devices below. This is a critical photolithography step.



1st Interconnect Layer

Dry Etch - Oxide

Contact Etch: A carefully designed RIE etch using fluorine chemistry to achieve vertical sidewalls.

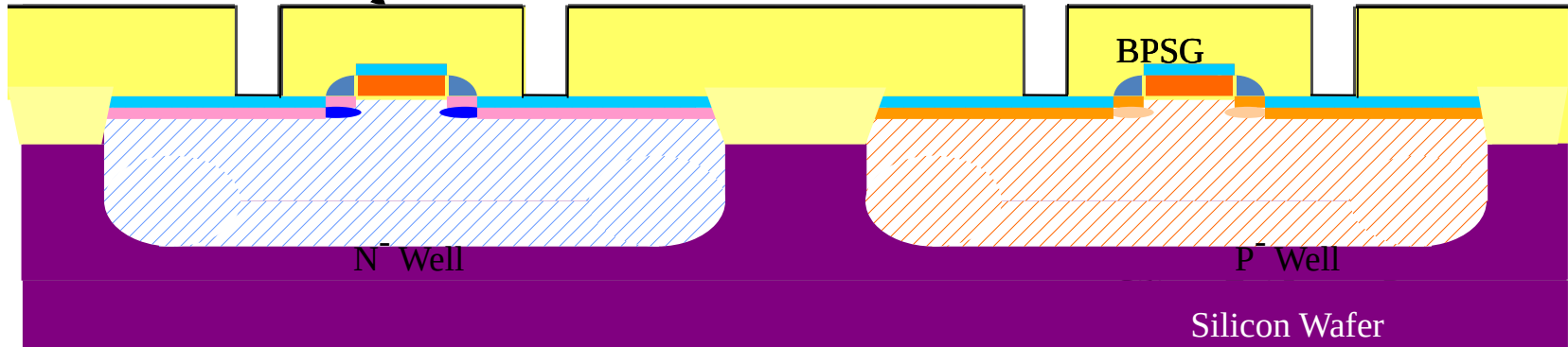


1st Interconnect Layer

MOCVD

Titanium Nitride Deposition: A sputter tool is used to deposit Ti and TiN. This layer will help the subsequent tungsten layer to adhere to the oxide.

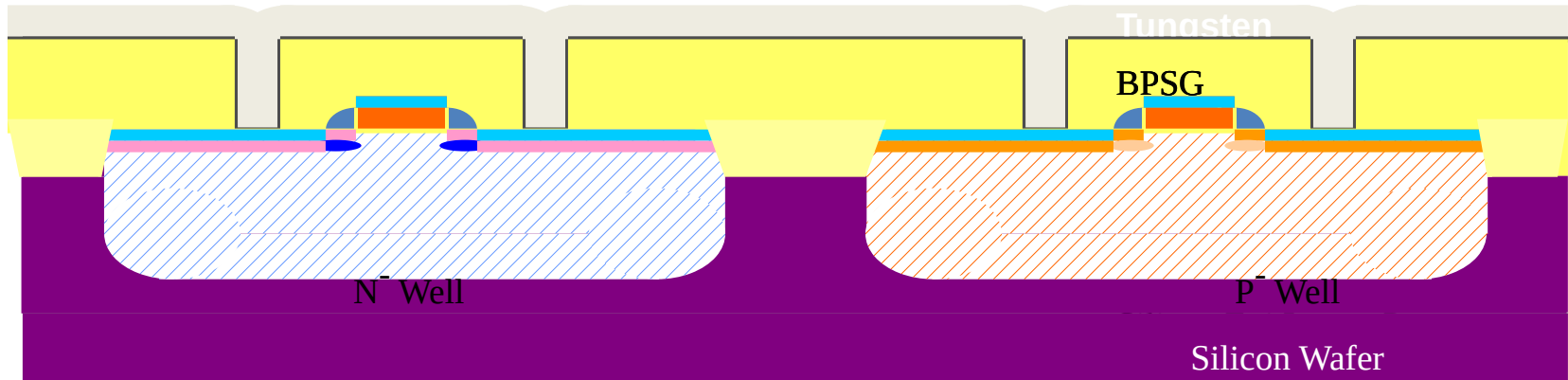
Ti+Titanium Nitride



1st Interconnect Layer

Tungsten CVD

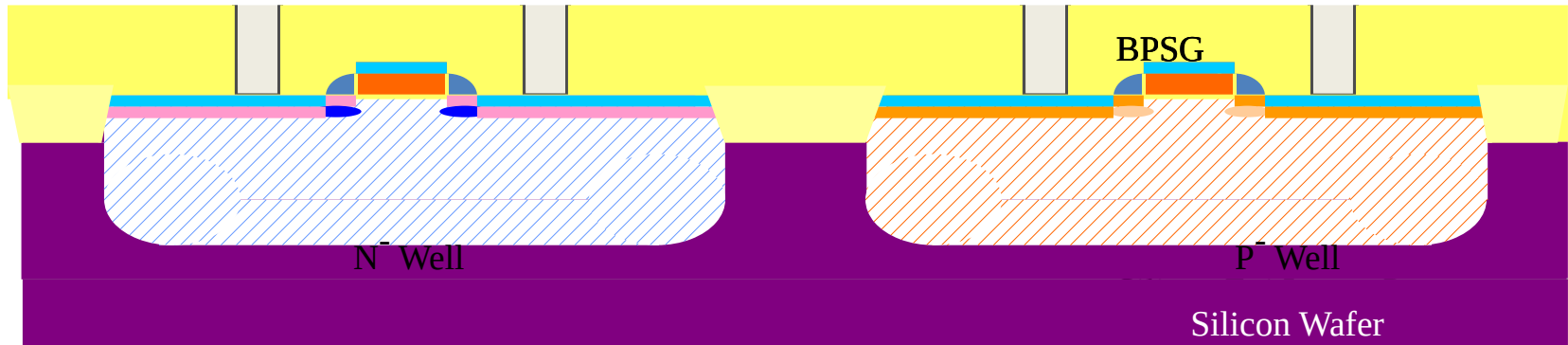
Tungsten Deposition: Tungsten is chosen because it deposits conformally (via CVD) and can fill the contact holes.



1st Interconnect Layer

W CMP

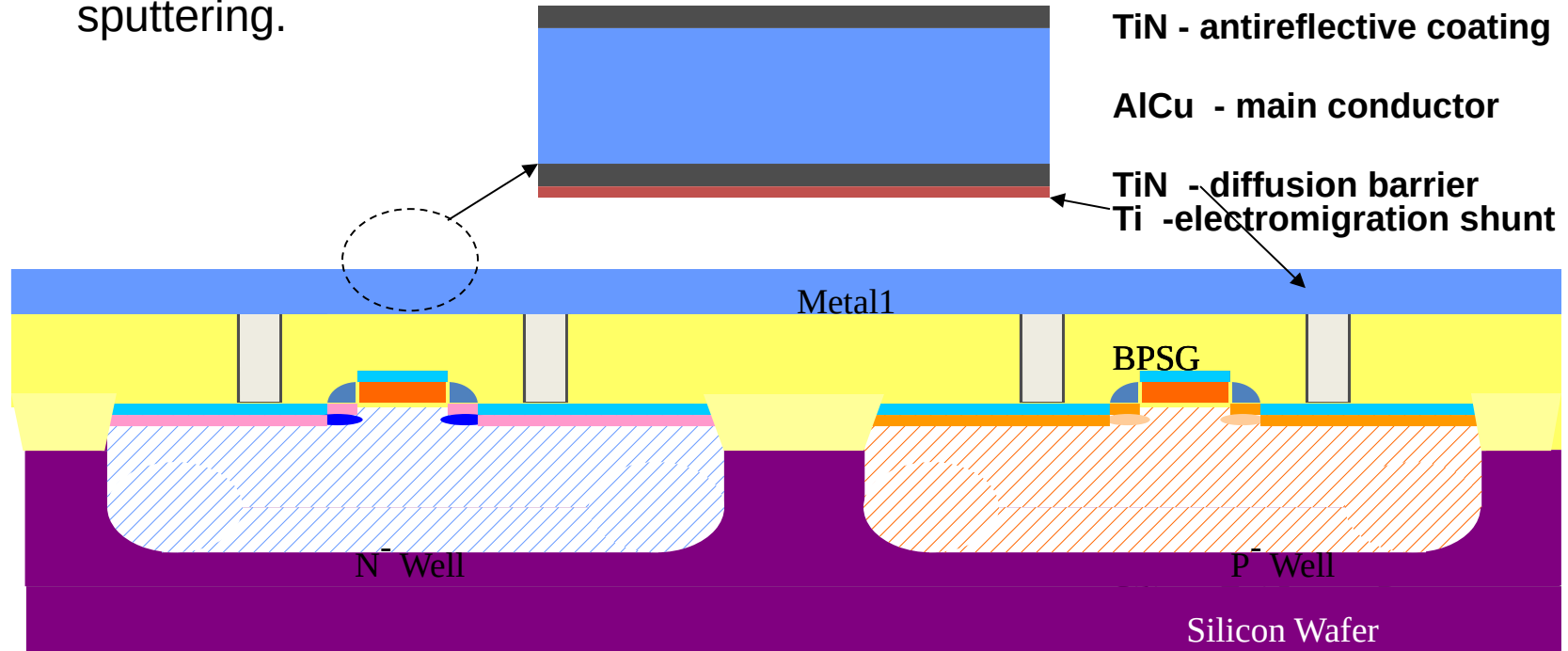
Polish Tungsten: CMP is used to remove the surface tungsten. The remaining tungsten forms “plugs”. The surface titanium nitride is also removed.



1st Interconnect Layer

Sputter

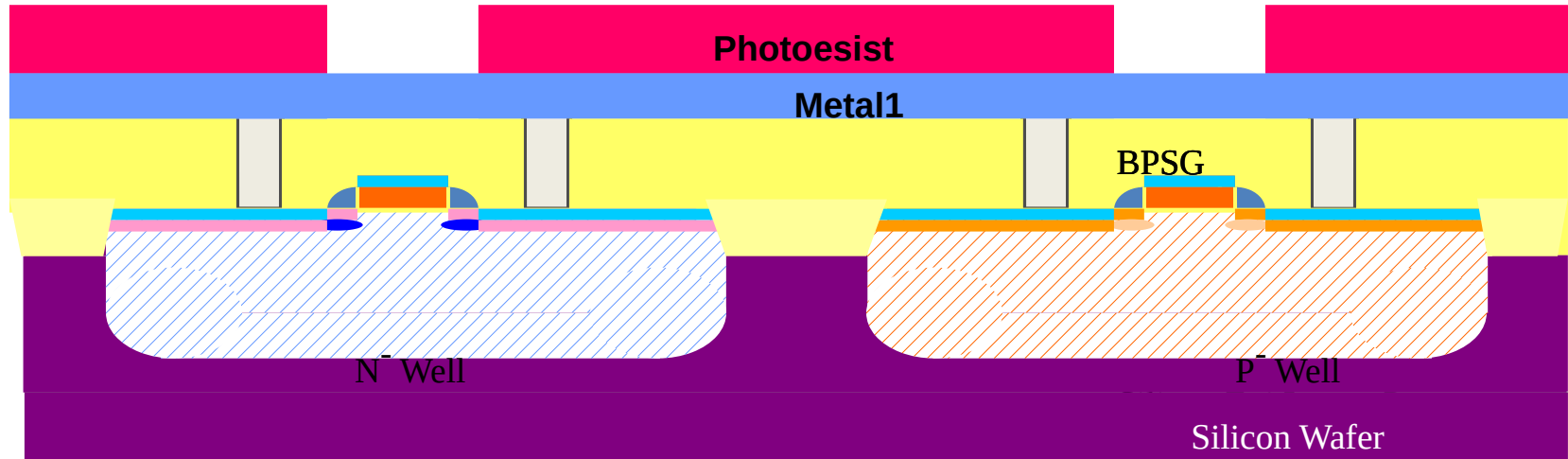
Deposit Metal1 Layer: Each interconnect layer is actually a sandwich of different layers. A sample is shown below. The films are deposited by sputtering.



1st Interconnect Layer

DUV Track & DUV Scanner – Patterning

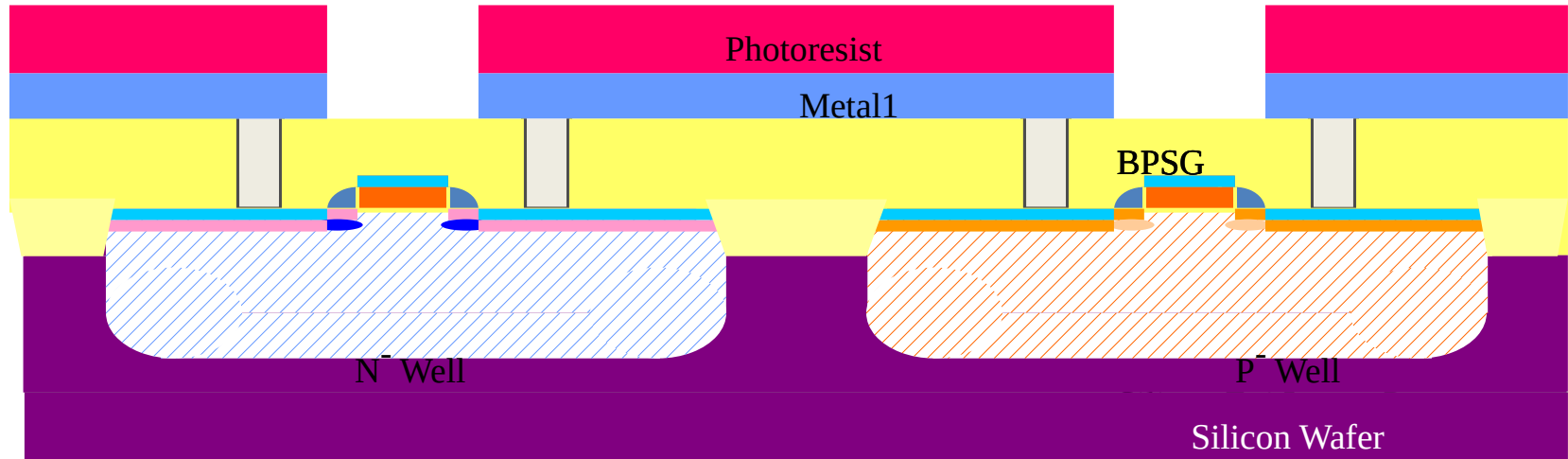
Pattern Photoresist for Metal1 Interconnects



1st Interconnect Layer

Dry Etch - Metal

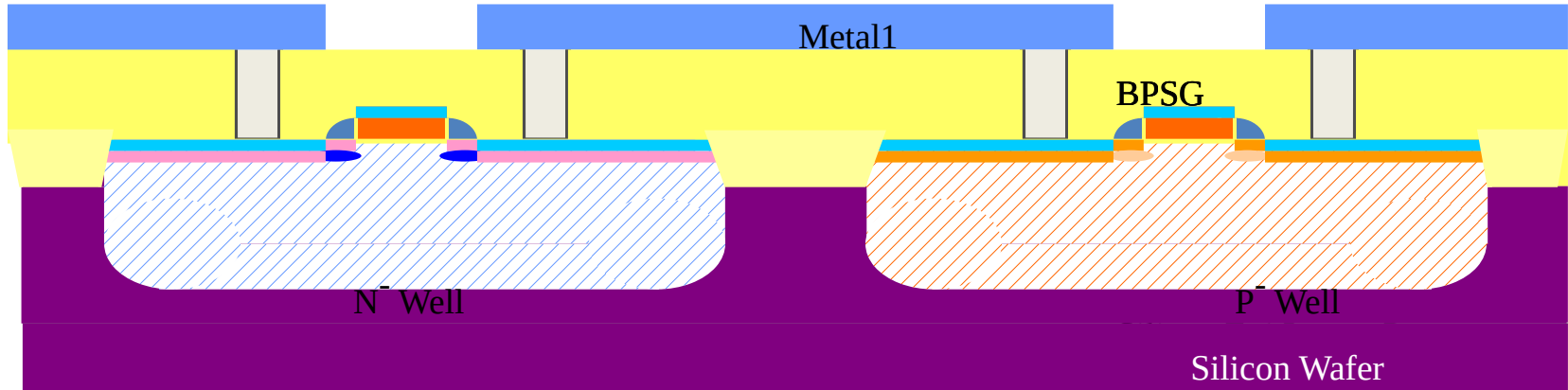
Etch Metal1: An RIE etch utilizing chlorine chemistry. Multiple etch steps are required due to the multiple different metal layers.



1st Interconnect Layer

Ash & Wet Solvent Strip

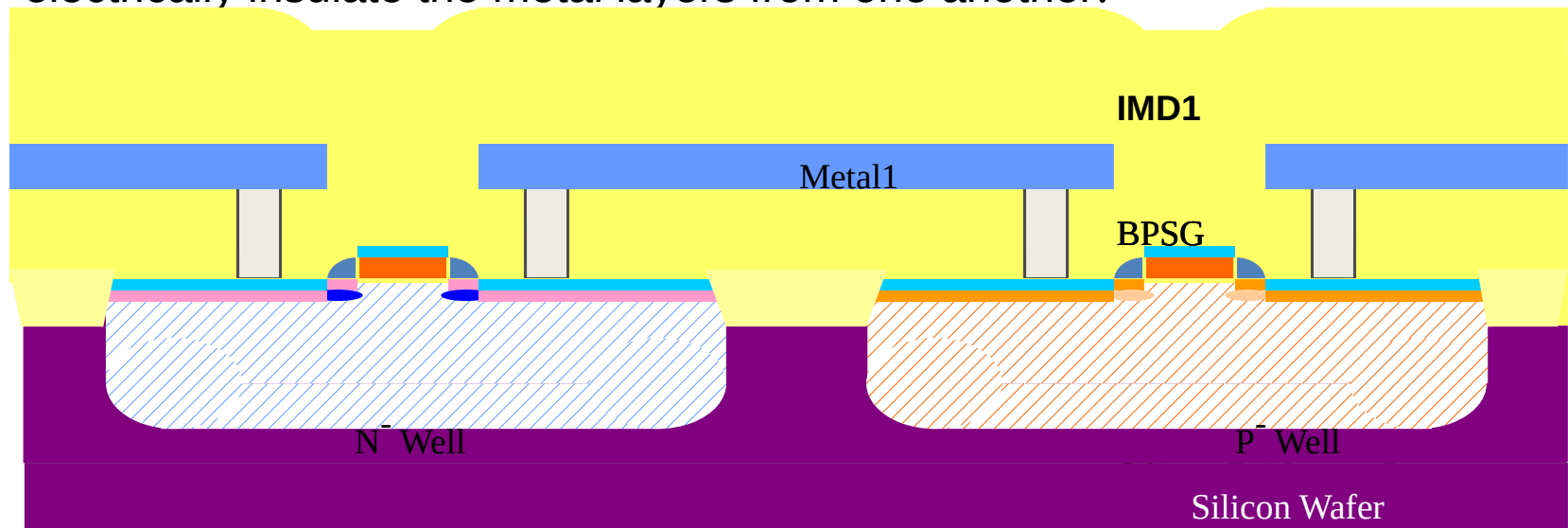
Strip Photoresist: First interconnect layer is completed.



2nd through Nth (as many as 6) Interconnect Layers

HD-CVD & CVD TEOS

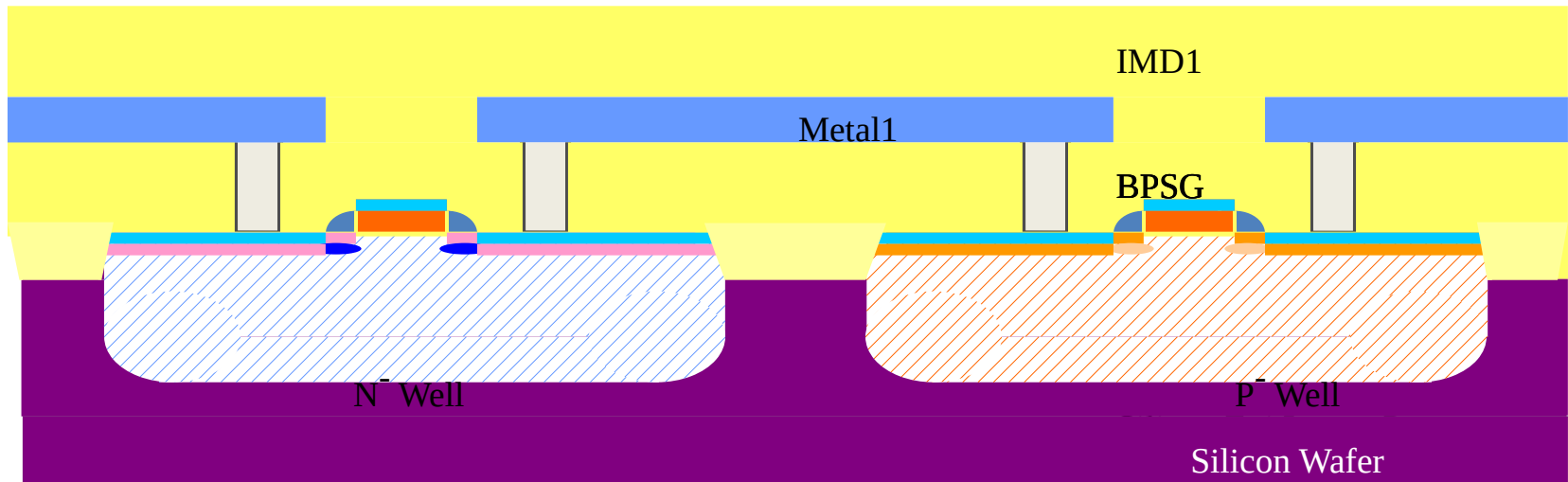
Deposit IMD1: Undoped silicon dioxide is deposited using successive CVD depositions to achieve filling between metal lines. Firstly USG deposited by HDPCVD after that TEOS oxide is deposited by CVD. This layer will electrically insulate the metal layers from one another.



2nd through Nth Interconnect Layer

Oxide CMP

Polish IMD1: Use Chemical Mechanical Polishing to achieve a flat surface on the IMD1 layer. If not removed, the bumps on the surface from the underlying topography would cause a problem for the subsequent



Contacts, Interconnects

- VIA-1 mask DUV
- VIA 1 etch/PR ash
- VIA 1 barrier Ti/TiN deposition
- W-fill
- W-CMP
- M-2 (Al) or Metal stack (Ti/TiN/Al/TiN) deposition
- M2 mask DUV
- M2 etch

Contacts, Interconnects

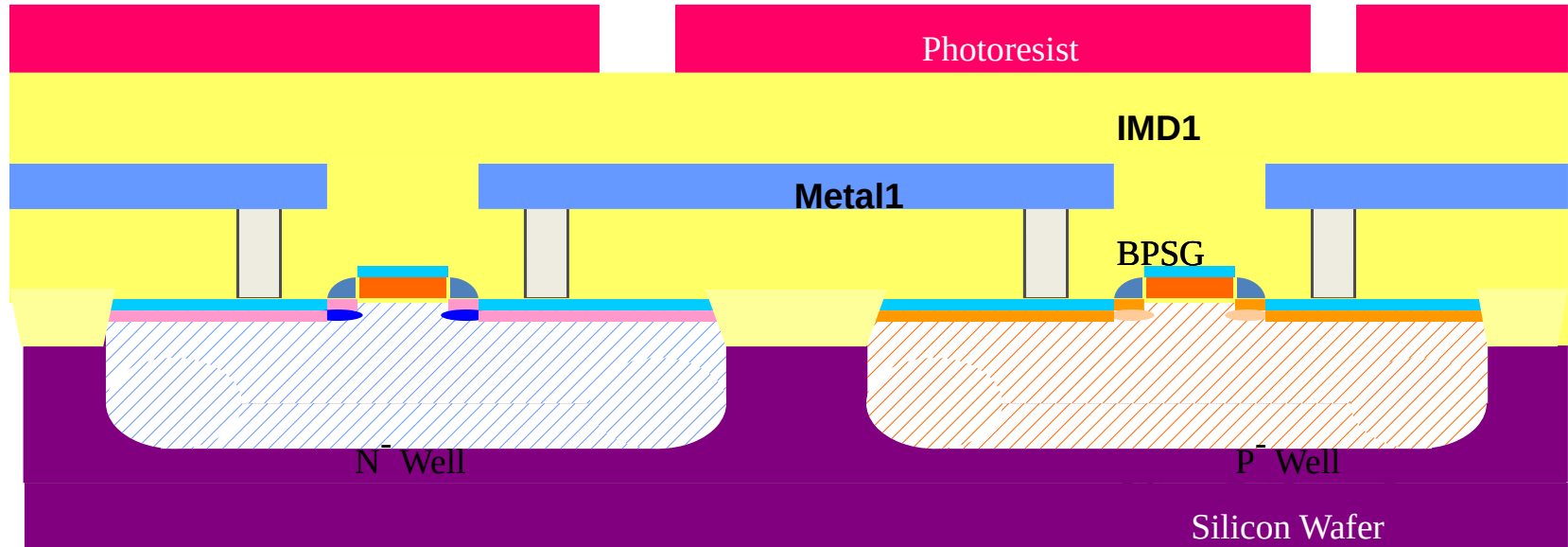
Repetition of stages from metal layers up to M5 etch

- Passivation oxide 1 deposition HDPECVD
- Passivation Nitride deposition PECVD
- Pad mask MUV (i-Line)
- Pad etch
- Nitride etch
- Oxide etch
- Al-Alloying/sintering
- Back side processing

2nd through Nth Interconnect Layer

DUV Track & DUV Scanner – Patterning

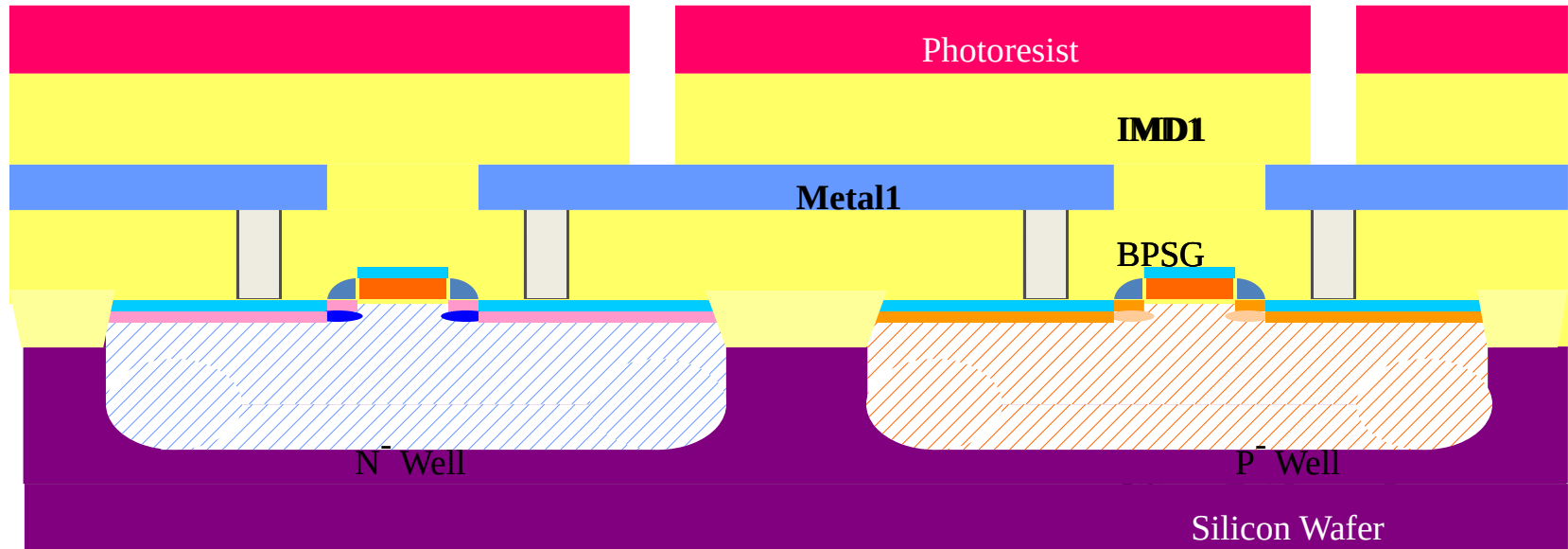
Pattern Photoresist to Define Vias: Vias are contact openings in the IMD layers enabling electrical access between metal layers.



2nd through Nth Interconnect Layer

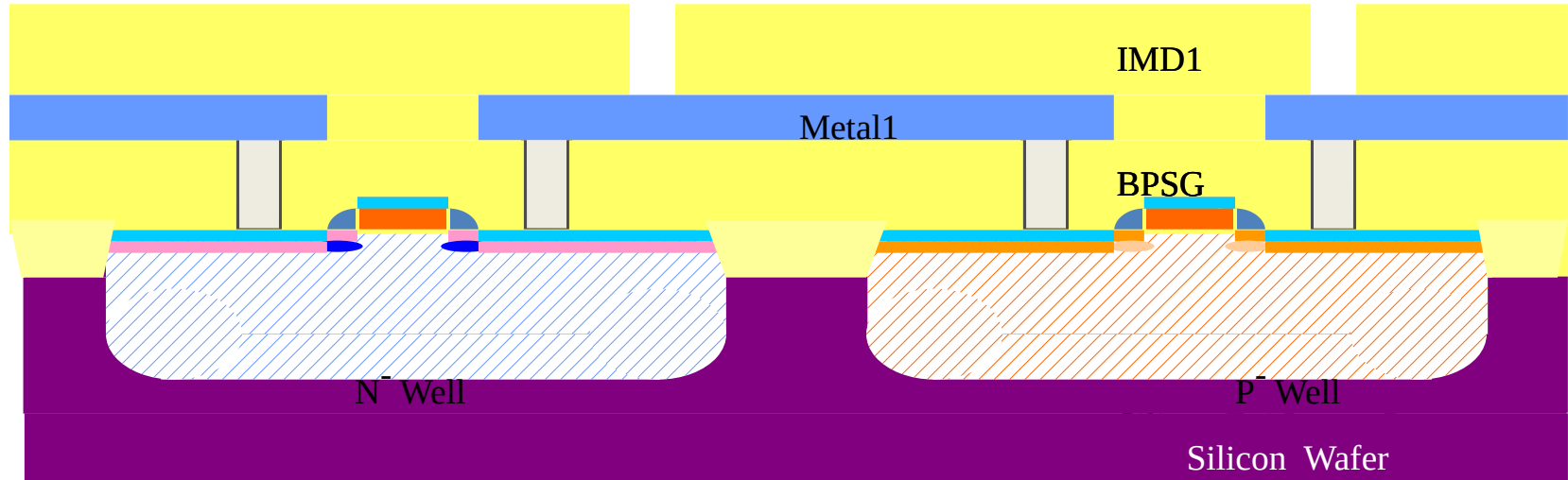
Dry Etch - Oxide

Via Etch: A carefully designed RIE etch using fluorine chemistry to achieve vertical sidewalls.



2nd through Nth Interconnect Layer

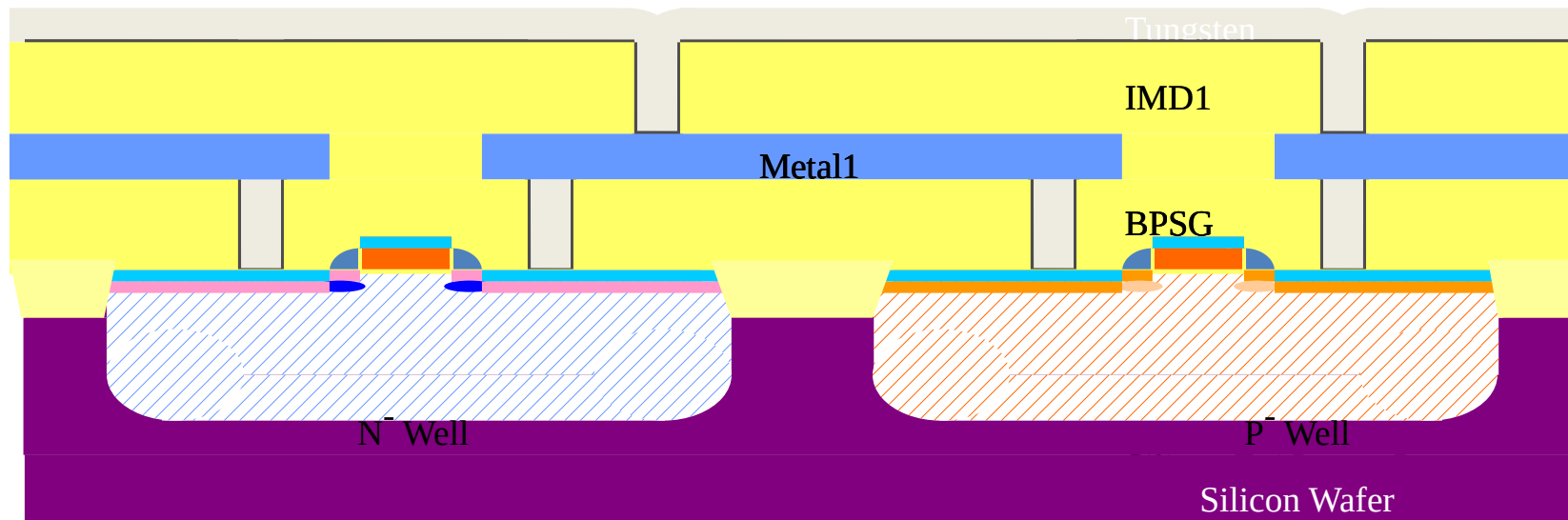
Ash & Wet Solvent Strip



2nd through Nth Interconnect Layer

MOCVD Ti & CVD Tungsten

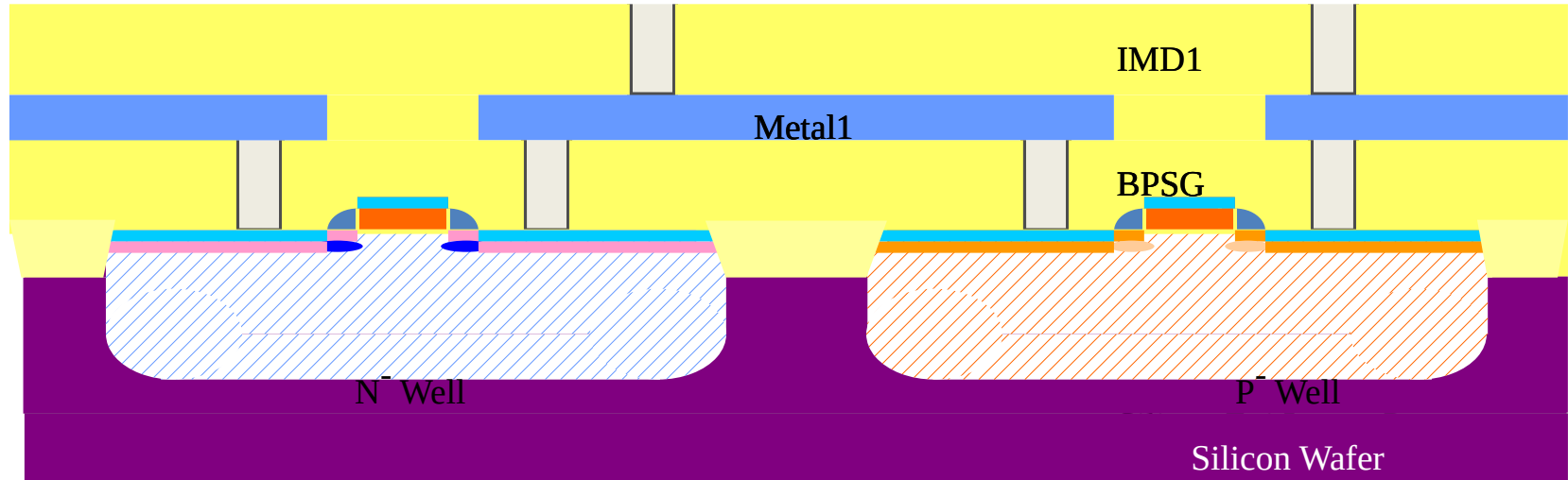
Deposit Titanium Nitride and Tungsten: Same as for the first interconnect layer.



2nd through Nth Interconnect Layer

W CMP

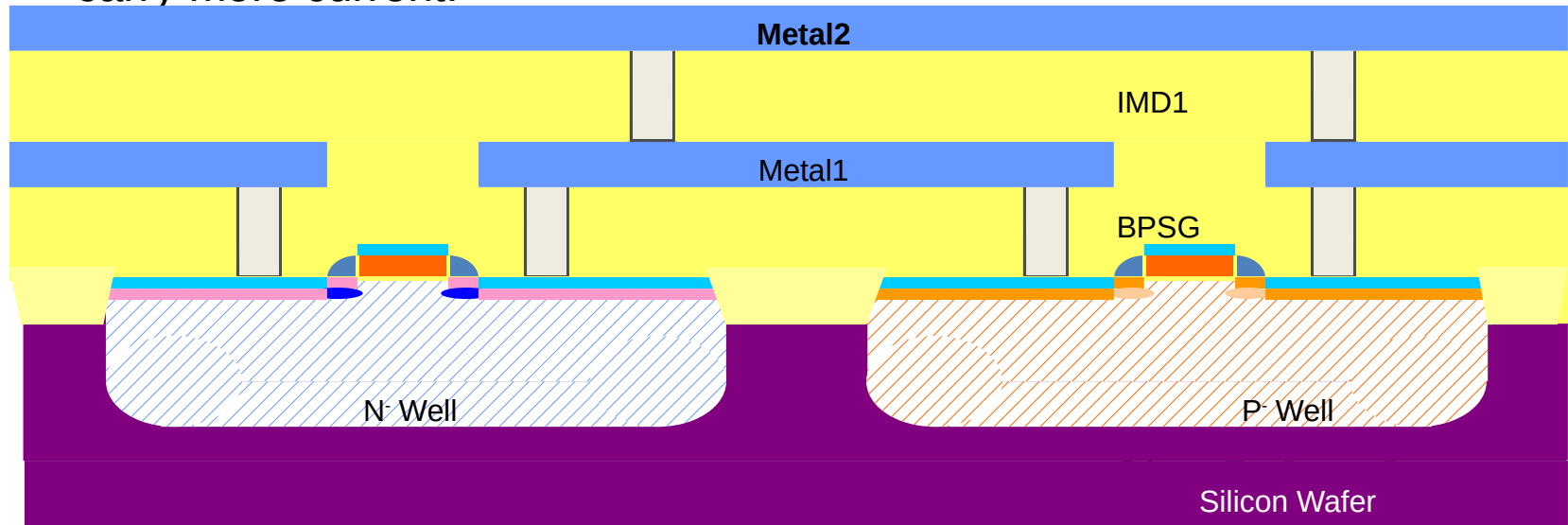
Polish Tungsten



2nd through Nth Interconnect Layer

Metal Sputter

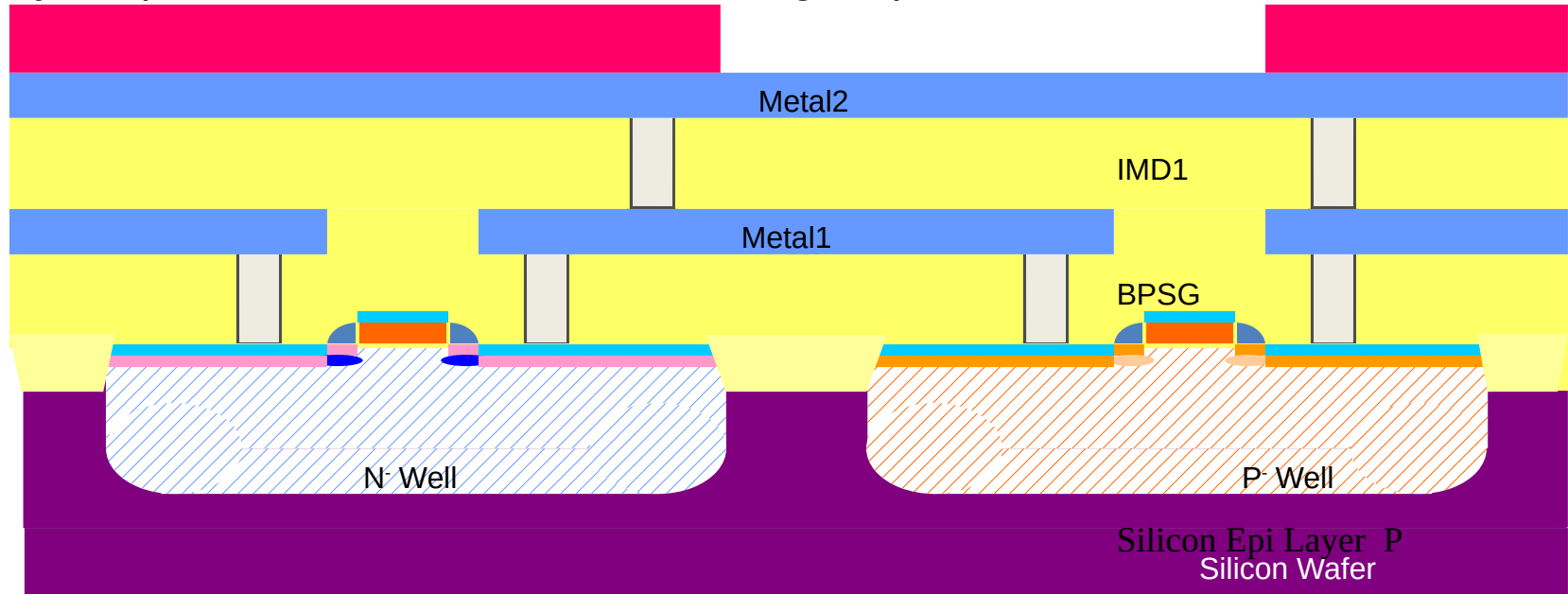
Deposit Metal2: Subsequent metal stacks will be similar to Metal1 but tend to increase in thickness and width as their runs are longer and they carry more current.



2nd through Nth Interconnect Layer

DUV Track & DUV Scanner – Patterning

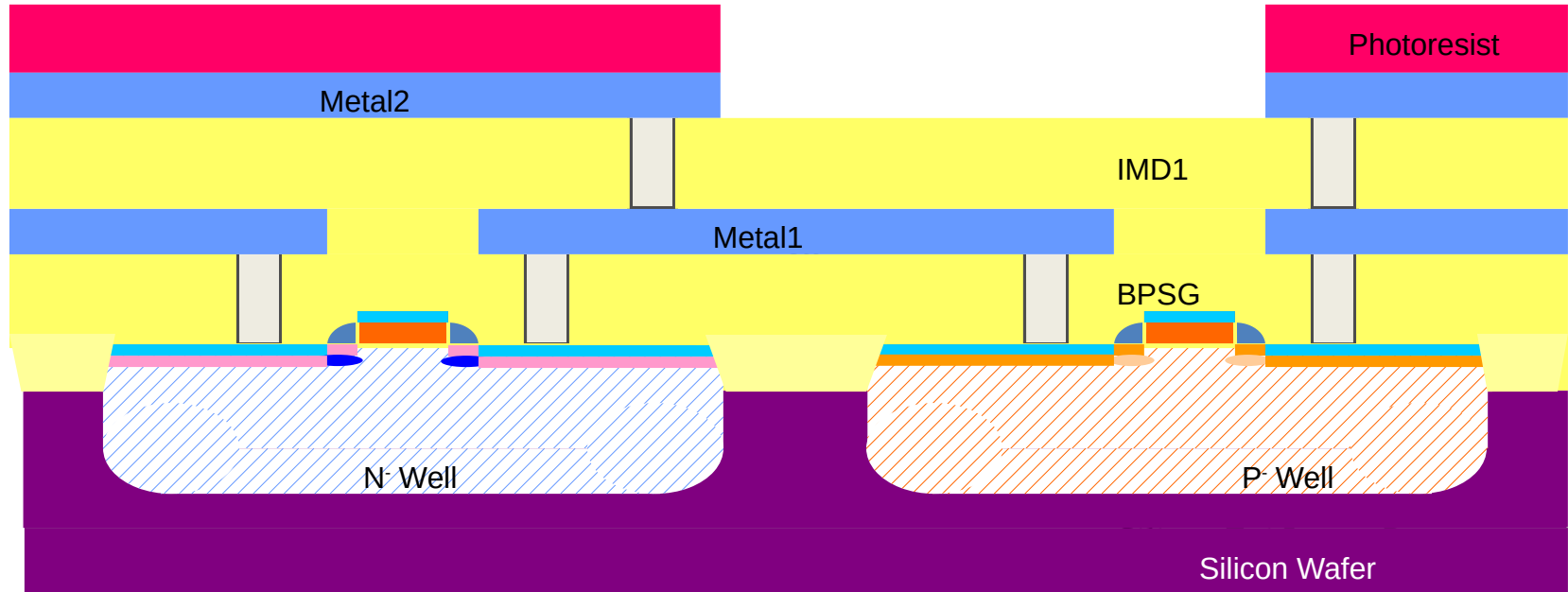
Pattern Photoresist for Metal2 Interconnects: Adjacent metal layers are patterned perpendicular to each other to minimize inductive coupling between layers (not shown in cross-sectional diagram).



2nd through Nth Interconnect Layer

Dry Etch - Metal

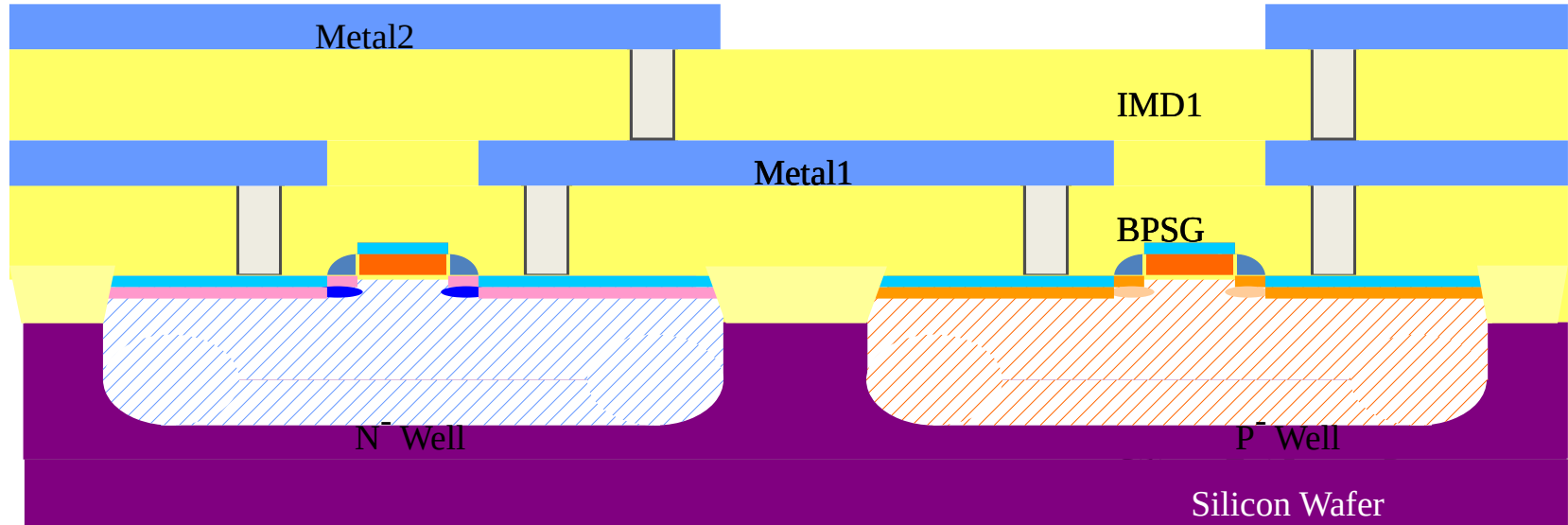
Etch Metal2: An RIE etch utilizing chlorine chemistry. Multiple etch steps are required due to the multiple different metal layers.



2nd through Nth Interconnect Layer

Ash & Wet Solvent Strip

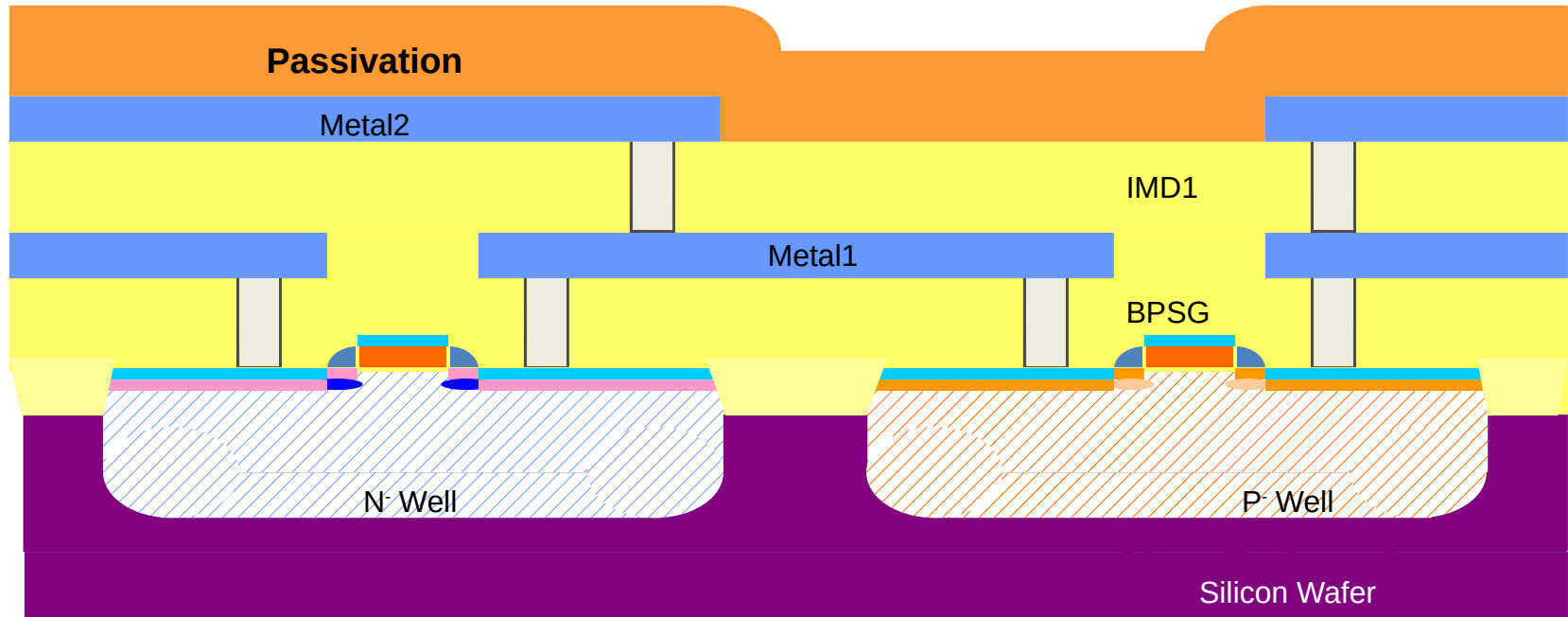
Strip Photoresist: 2nd interconnect layer is completed.



Passivation

HD CVD – Deposition

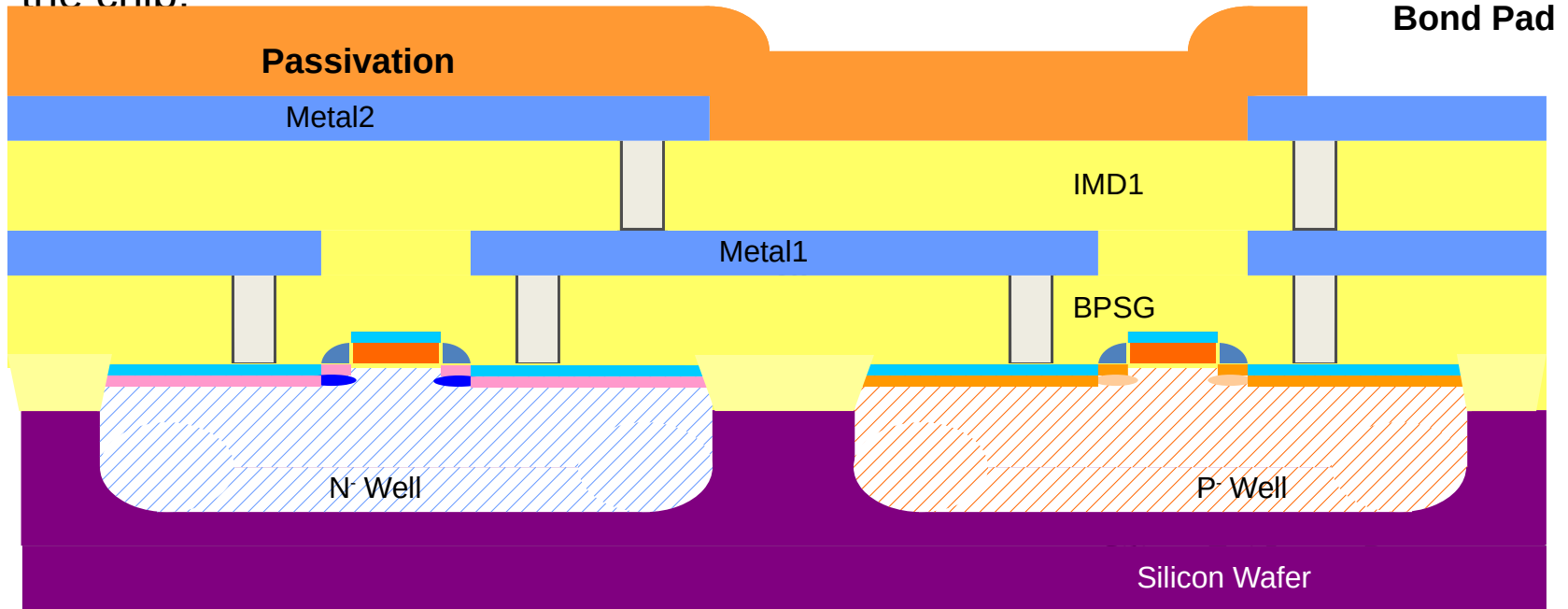
Deposit Passivation Layer: There are many types of passivation (silicon nitride, silicon oxynitride, polyimide, and others). Its purpose is to protect the completed circuit from scratches, contamination, and moisture.



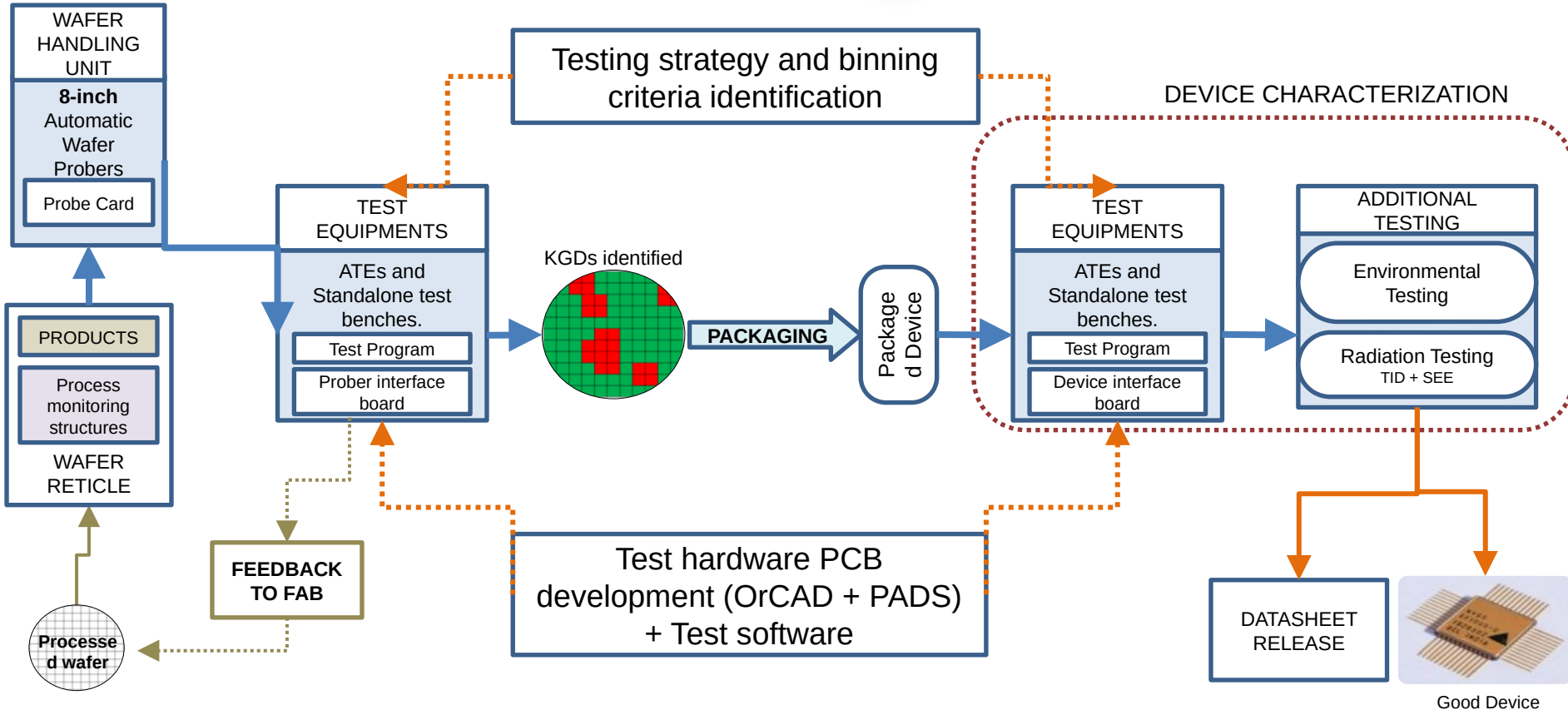
Passivation

Dry Etch – Oxide Etching

Pattern Passivation Layer: Bond pad openings allow electrical access to the chip.



VLSI Testing Flow



Test Capabilities

AUTOMATIC TEST EQUIPMENTS



Test Capabilities

ENVIRONMENTAL TESTING

- Temperature Forcer systems: Range: -60°C to $+150^{\circ}\text{C}$ Accuracy $\pm 1^{\circ}\text{C}$
- Environment Chamber: Range: -180°C to $+150^{\circ}\text{C}$ Accuracy $\pm 1^{\circ}\text{C}$
- Temperature chamber: Range: -55°C to $+150^{\circ}\text{C}$ Accuracy $\pm 0.5^{\circ}\text{C}$



OVERHEAD DOCKING

VLSI Assembly & Packaging Process Flow

VLSI Assembly & Packaging is MIL-PRF-38535 compliant facility operating in Class 100 Clean area to cater High Reliability devices with below equipments:

1

• WAFER DICING-

- Mechanical Dicing up to 12 inch wafer size, twin blade dicing for step cutting
- Can cater different substrates up to 2.4mm thickness.

2

• DIE ATTACH & ADHESIVE CURING-

- Manual die attach operation with space grade adhesives on Ceramic substrates with DIP, QFP, SOIC and PGA outlines having size variation from 5mm lengths to 150mm lengths.
- RTC Belt furnace: IR heating Multizone furnace; Multiple Curing profiles; Maximum temperature of 800°C and belt speed of 40ipm.
- Oven: 200 liter volume with multiple storage levels; Multiple Curing Profiles; Max Temp of 300°C.

3

• WIRE BONDING-

- Wire bonders (2nos): Semi-Automatic Ultrasonic wedge fine pitch bonder; Capability of minimum die pad bonding of 57 micron x 57 microns at pitch of 65 microns with 25microns Al wires.

4

• HERMETIC SEALING-

- Au-Sn Soldering, Glass and Epoxy Sealing: Sealing solutions for ASICs and Imagers; Manual operation with sealing lids varying from 10 mm lengths to 110 mm lengths.
- Belt furnace: Multi-zone heating belt furnaces; Multiple Sealing profiles; Maximum temperature of 900°C and belt speed of 12ipm.

5

• MARKING & TRIMMING-

- Semi-Automatic Laser marking machine for ceramic, plastic and metal packages; Max power 30 W; minimum character height 0.2mm with Logo and ESD stamps through dxf file inputs.
- Forming & Trimming machine: Adjustable Quad Flat Pack Cut & Forming for DIP, QFP and SOIC outlines. 8K per month (for 120Pin CQFP equivalent device)

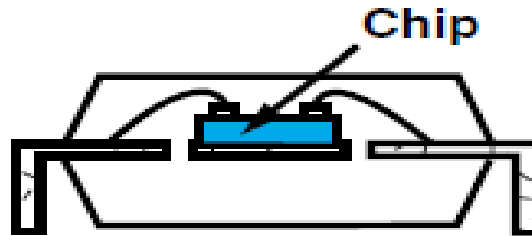
6

• VISUAL & NDT INSPECTIONS-

- Universal Bond Pull Tester: In-line wire bonds integrity and die attach strength evaluation; Wire bond pull 100gmf; die shear tool module 200Kgf; Tweezer Pull 10kgf.
- High Accuracy Measuring Microscope (2nos): Die and Pre-Cap Inspections; Max span 250mm x 250mm; Upto max zoom of 1000X.

What is Packaging?

It is the interconnection of electronic components and the establishment and regulation of their operational environment in order to meet specified performance, reliability and cost requirement.

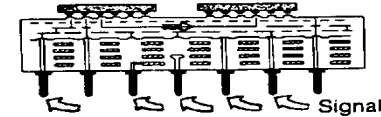


Purpose of IC Packaging

➤ SIGNAL DISTRIBUTION

- Reduces the cross-talk , reflections and other distortions during the transmission of signal .
- Use of best electrical conductors to reduce noises.

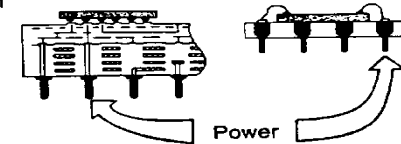
Signal Distribution



➤ POWER DISTRIBUTION

- It reduces the switching noise
- By using material with highest electrical conductivity(for lowest voltage drop).

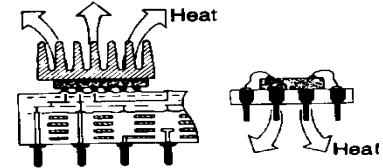
Power Distribution



➤ HEAT DISSIPATION

- By using heat spreaders

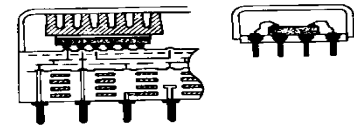
Heat Dissipation



➤ PACKAGE PROTECTION

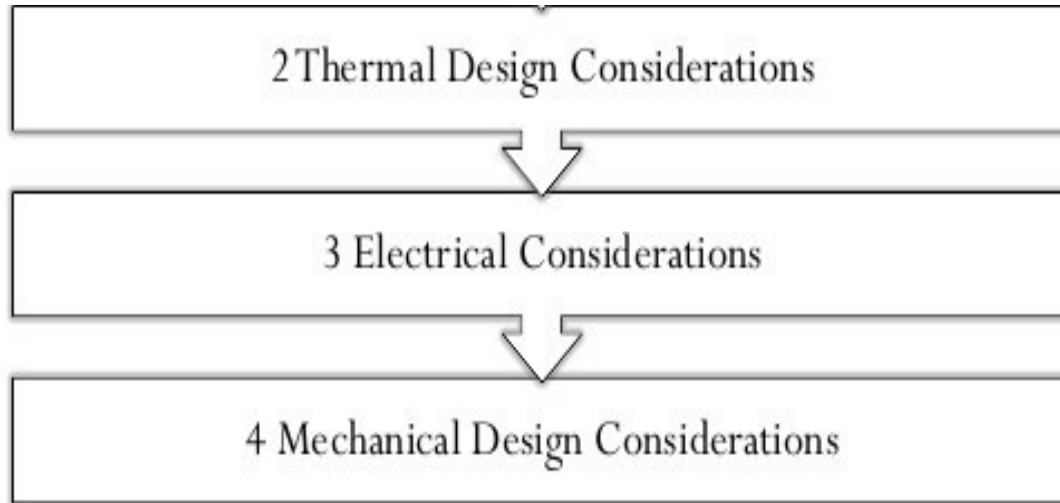
- Physical support(From vibrations or breakage)
- Chemical protection(From corrosion)

Package Protection



➤ MAINTAINING OPERATION OF IC

DESIGN CONSIDERATIONS



IC PACKAGING MATERIAL

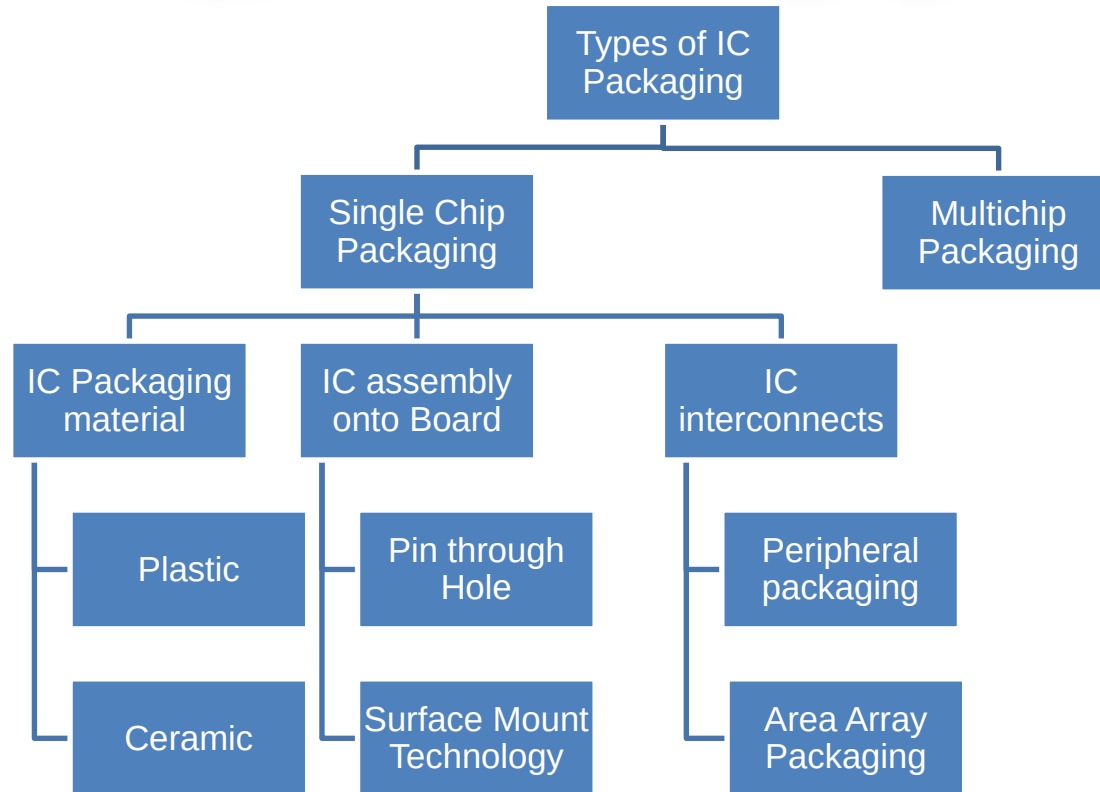
PLASTIC

- Less costly but high-thermal resistance
- Absorb moisture
- Hygroscopic nature

CERAMIC

- Chip placed in cavity and bonded to conductors
- Sealed against the environment
- High permittivity
- Expensive

Types of IC Packaging

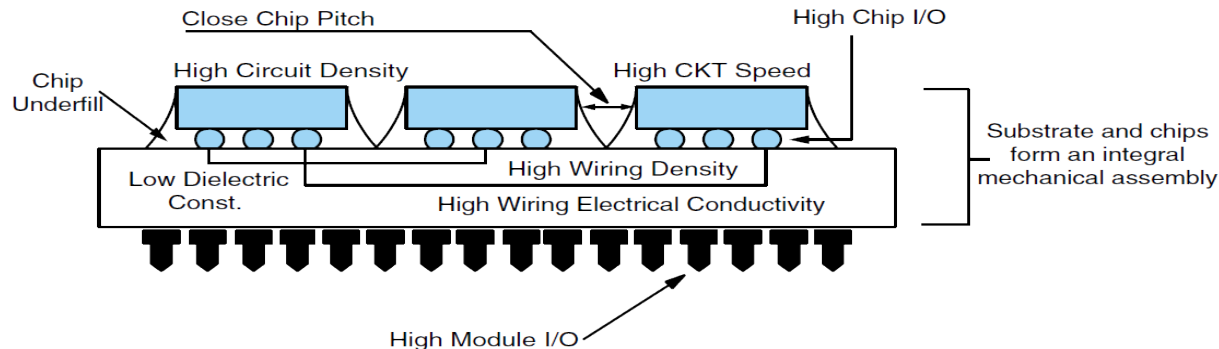


MCM - advantage

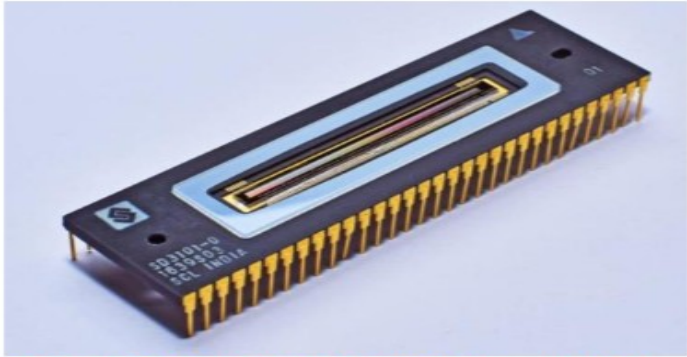
- **Improved Performance:** closer positioning of the dies on the substrate and shorter interconnection lengths should enhance system speed dramatically.
- **Higher Integration Density:** substituting several packages for one slightly larger but single package, should either free board real estate for other use or help reduce the board size.
- **Lower Power Consumption:** smaller drivers are needed resulting in lower power consumption.
- **Mixed Signal Applications:** MCP allows the integration of chips made from different technologies in one package.
- **Lower Cost:** cost savings result from fewer packages with a fewer number of leads, a simplified board layout, and the feasibility of mixed technologies in the same package.

4 major Principles:

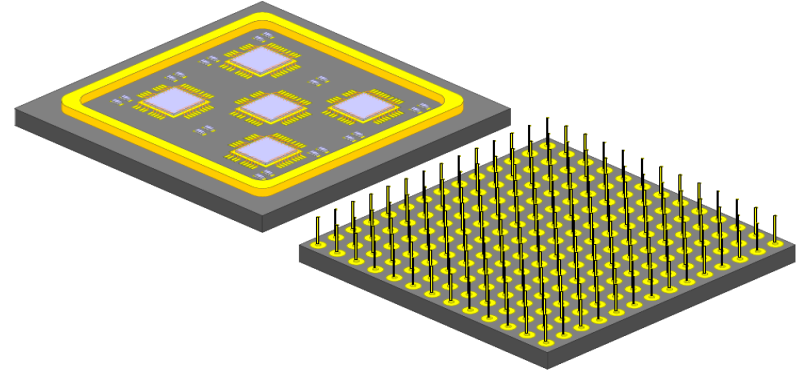
- Signal I/O
- Thermal mgmt
- Mechanical support
- Environment Protection



MCM – sample design cycle

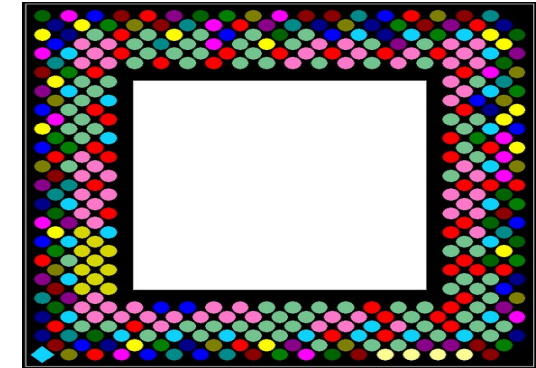
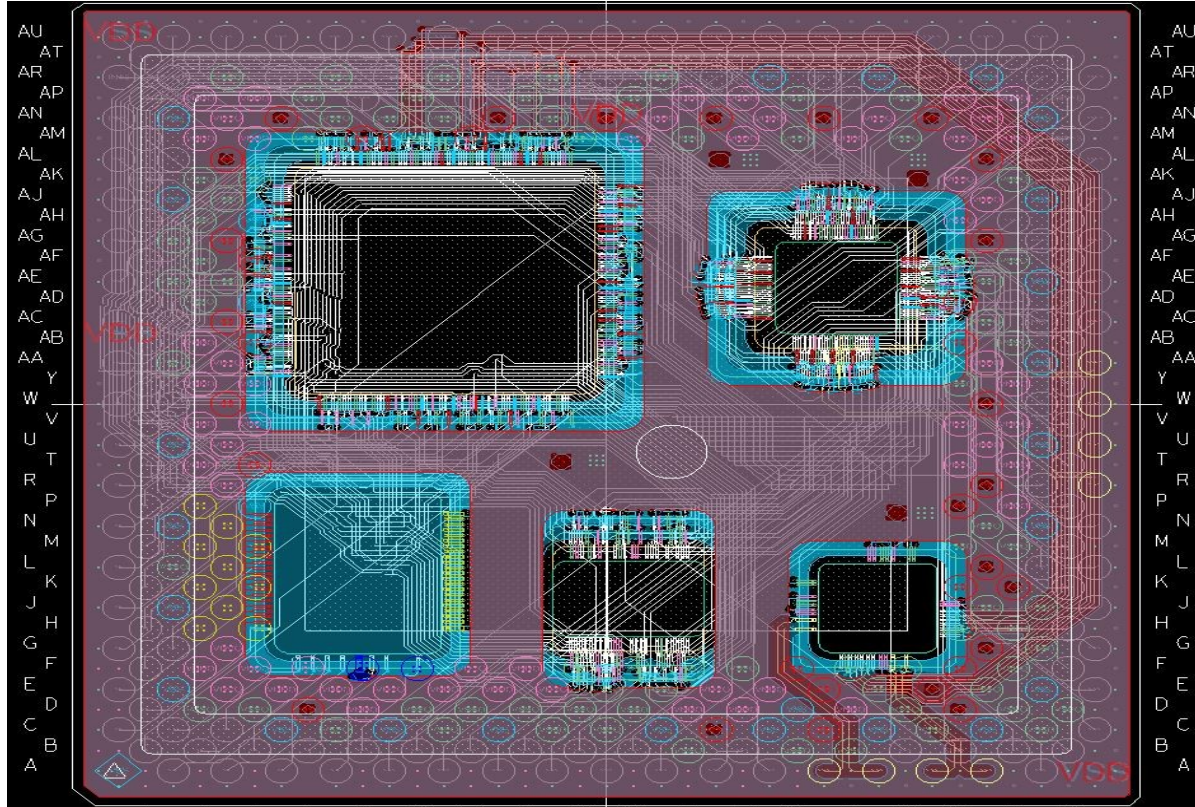


Single chip



Multi – chip module

MCM- package – layout



PGA pin out

Metallisation in different layers across the thickness